

Toward a topological theory of the aspectual conceptualization of events

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Master's thesis (M2 Logos)

Université Paris Cité

September 6, 2024

MASTER LOGOS



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Abstract

I present a mathematically intelligible theory of event conceptualization, known as lexical aspect within linguistics, based on Croft's (2012) diagram-based account. Croft defined aspectual types using sequences of qualitative states, giving full status to the qualitative component of the unfolding of events. I make this idea formal using objects from general topology, by defining each aspectual type as the conjunction of a qualitative topological space, together with a continuous trajectory within that space. The topology on the space only distinguishes qualitative states inasmuch as they differ with respect to the unfolding of the event, discarding otherwise irrelevant distinctions between event types. Continuous transformations can change one space into another by collapsing distinctions, regimenting the shifts in granularity proposed by Croft for aspectual coercion. This preliminary, more constrained framework for event conceptualization already allows us to evaluate existing aspectual distinctions and operations, and conjecture new ones. In light of this theory, I differ from Croft in my treatment of semelfactives, arguing that they are telic predicates with a special activity reading, similar to Rothstein (2008), who proposed semelfactives were activity predicates with naturally atomic minimal events. I also deny aspectual status to nonincremental accomplishments (*solve the puzzle*) and irreversibility, suggesting that the proposed diagnostics are sensitive to other inference-making mechanisms. Finally, I conjecture transformations to explain the effect of aspectual verbs, which could conceivably help formulate an algorithm for computing the presuppositions of these soft triggers, based on the aspectual representations provided by the theory.

Acknowledgments

Foremost, I wish to thank my supervisor, Salvador Mascarenhas, for making this project possible in the first place. I was lucky to meet him at a pivotal time in my studies, and feel truly grateful for his intellectual and human generosity, which provided the soil and water for the ideas behind this thesis to develop. I hope to come closer to the breadth and rigor of his thinking as a researcher.

This work owes a tremendous intellectual debt to Bill Croft, whose elegant work was just begging to be made formal. He welcomed the effort with kindness and important, thoughtful remarks, which I very much thank him for. I also owe much to Jonathan Ginzburg, both as a teacher and as a committee member, for his original, inspiring view of semantics. Thank you to Michael Goodale and Sasha Lanine, for engaging and commenting on my work critically.

For the rest, and getting me out every so often, the usual suspects: Émile and Tossia, Charles, Matéo, Lucas, Noah, Petit Gaga, Rémi, Elodie, les Huns, and my wonderful family, who support me in all the important ways.

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1. Introduction

Linguists have long noted that there can be “different ways of viewing the internal temporal constitution of a situation” in the same language (Comrie 1976): the same intuitive goings-on in the world may be presented as unfolding over time differently when they are put into words (Croft 2012). The conceptual–grammatical categories behind these phenomena are collectively called **aspect**. For instance, in English, the presence of a conceptual bound on the event affects the grammar of duration, as in (1a) and (1b) (inspired by Vendler 1957).

- (1) a. Kim ran for five minutes.
b. Kim ran a mile in five minutes.

As Krifka (1998) noted, the two descriptions are clearly compatible, but (1b) asserts that the situation reached some **completion** (also called ‘terminus’ or ‘result state’). Meanwhile, no such completion is represented in (1a). Predicates with (without) a completion are called telic (atelic), and the sensitivity of grammar to telicity is reflected in the duration adverbials used for either type, as above.^a In examples such as this one, speakers display striking systematicity in how they present events using the grammatical tools of their language. Yet, the factors determining different predicates’ behaviors, and the problems of composition and coercion, are still ill-understood. This makes the study of so-called *lexical aspect* a notoriously slippery domain, with fluctuating classifications and analyses (Bache 1995; Klein 1994; Moens and Steedman 1988; Smith 1997; Vendler 1957), and precious few formal theories – van Lambalgen and Hamm (2005) being a notable exception. I aim to raise awareness about this gap and to provide insights from mathematical theories on how it might be filled, building on a cognitive-linguistic account by Croft (2012).

In keeping with the remarks above, Croft (2012) convincingly argued that the aspectual difference between predicates is foremost one of contextual **conceptualization**, or **construal**: to express an experience in language, one must superimpose a certain semantic structure on it, a construal, above and beyond its matter-of-fact properties. Different construals may involve anything from different participants to framing the same action against different conceptual backgrounds (as in the running example above), or attending to a different phase of the action (see Example (4) below). This approach generalizes the notion of “viewpoint,” often used for *grammatical aspect*, to lexical types themselves, so that construals are no longer an inherent property of predicates.^b For this reason, I refrain from using the term ‘lexical,’ but keep the useful term ‘aspect,’ to refer to phenomena relating to the temporal conceptualization of events. In fact, although Croft (2012) himself keeps linguistic representation separate from more general conceptual representations (pp. 16–17), it is entirely conceivable to push the view further, and adopt as a working hypothesis that **aspectual phenomena could follow from a prelinguistic event conceptualization faculty in humans**. Then, if event construal is required at a deeper cognitive level, linguistic facts can be used as windows into the mental structures underlying event conceptualization.

At any rate, linguistic aspect is surely relevant to the study of the human language faculty, and the categories it involves, including with respect to more traditional problems in semantics. Aspectual verbs like *start*, *stop* and *finish*, or even the English Progressive construction clearly manipulate the presuppositional and asserted content of the verb phrase they modify. While atelic predicates tolerate the Progressive without any appreciable difference in assertion, see (2a), a telic predicate in the

^aThough the mile in question could have been Kim’s goal in running, note that this is not necessarily the case: the event can be presented as having a completion irrespective of the agent’s intentions.

^bThough many predicates may have a statistically default construal, which is not actually regarded as basic (Croft 2012, p. 92).

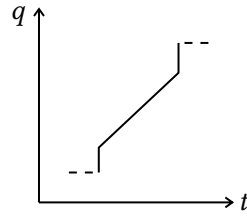


Figure 2: Representation for accomplishments, profiling the full transition phase

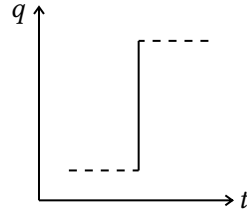


Figure 3: Coarse-grained representation for (3b)

Progressive will not assert the culmination, as in (2b). Despite this however, the possibility that Kim might complete the mile is still made salient, in contrast to (2a) which is compatible with but does not bring out that possible course of events.

- (2) a. Kim was running.
- b. Kim was running a mile.

Understanding which presuppositions are derived, which must be stipulated, and how they differ from salient possibilities such as in (2b), would have strong implications for human semantic competence. We believe lexical aspect brings a privileged domain in which to investigate these questions.

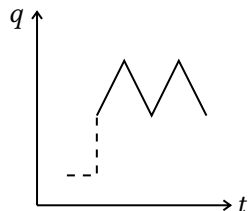
Croft's (2012) account of aspect suggests that salience and especially attention could play a major role in the conceptualization of events. In his system, each aspectual type is a construal, which involves *profiling* (highlighting) part of a path, which I call a trajectory, traced out by an entity within an **abstract qualitative dimension** q . This trajectory is represented in a two-dimensional graph, plotting qualitative state against time t and using a solid line for its profiled part. For instance, the usual construal for (1b) would be represented as in Figure 2, with a beginning stage, a transition stage and a result stage; with the second stage (the actual running) being profiled. This kind of construal is called an **ACCOMPLISHMENT**, i.e. an activity with a culmination. As far as aspect is concerned though, the actual modes of being represented on the y -axis should not matter, so we can even abstract their interpretation and only consider their **cohesion with respect to the construed unfolding of the event**. In this context then, any event construed as an accomplishment, like *Reda ate a plate of pasta*, would enjoy a three-part graph identical to Figure 2, because its construal belongs to the same aspectual type.

One crucial insight of Croft's is that construal can change with attention. In his representations, attention manifests itself in two distinct ways: first, the **collection of states** in the qualitative dimension is open to construal, according to the desired level of detail, or the relevant facet of the event. Though the duration of the event and its successive phases are clearly relevant in (3a), one may see the event as a mere point, as in (3b). The change from one level of detail to the other involves a more coarse-grained conceptual dimension, and a simpler associated trajectory, as in Figure 3.

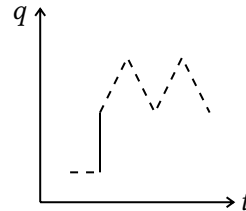
- (3) a. Kim ran a mile in five minutes.
- b. Kim ran a mile on August 9, 1995. (Croft 2012, p. 98)

Second, the trajectory has a **profiled subpart**, which changes with construal. Consider an undirected running event like running around in (4), with a zigzag line representation standing for repeated actions of a similar type (Croft 2012, p. 61). It can be construed either as an atelic activity in the progressive (left), or as the (telic) inception of that activity in the simple past, in the latter case with the same meaning as *start running around* (right). In Croft’s system, these alternatives simply differ in the profiled subpart of the trajectory (Croft 2012, p. 102).

(4) At noon, Kim was running around.



At noon, Kim ran around.



What different qualitative states are considered relevant, but also what parts of the entity’s path are brought under consideration, both vindicate a notion of attention.^c The idea is especially interesting, as it suggests that certain faculties of higher cognition could involve mechanisms of attention much like in the visual domain. Further, those mechanisms could background certain elements away from attention, just like happens to presuppositions among the class of possible entailments of a sentence; thus raising the possibility that the two phenomena might be related.

Despite its deep implications for event semantics, Croft’s (2012) account of event conceptualization is ultimately based on diagrams which **lack truly formalized, mathematically intelligible content**. Indeed, by his own account, Croft’s intentions were not to provide a logical calculus for his representations (p. 10). Unfortunately, this unregimented nature of the q dimension means that several of its important features are only intuitively defined, and left open to interpretation. Its points evidently change depending on the aspectual type, but this is nowhere represented in the diagrams (Croft 2012, p. 54): either two points might allow instantaneous jumps between them (Figure 3), or they might have an intervening continuum of points (Figure 2). A very insightful aspectual operation in the system, change of granularity, is only described precisely case by case instead of in a general setting, thereby lessening its explanatory power. Here, I aim to **restate the definitions for aspectual types in a mathematically precise and rigorous way**, while salvaging the insight about attention. For this, I focus on conceptual spaces and use ideas from **general topology**, the study of spaces up to deformation, to cash out the notion of cohesion between different modes of being suggested by Croft’s representations for aspectual types. Tools from this domain of mathematics illuminate Croft’s informal qualitative dimension, both in how an entity can navigate it, but also in its manipulation to yield alternative construals. For instance, a collection of two points can be chosen to be more or less cohesive to yield different spaces, each allowing a more or less restrictive class of trajectories within it. While an entity cannot move between the isolated points in the **DISCRETE SPACE** of Figure 4a, there is another two-point space, the **SIERPIŃSKI SPACE**, whose points are stuck together but remain meaningfully different. This space allows certain kinds of instantaneous transitions between the two points, just like in Figure 10, giving us a regimented notion of quantum jump as needed for the punctual change of Figure 3. This contrasts with the last, least discriminate kind of two-point space, which does not feature

^cSimilar operations are available with Moens and Steedman’s (1988) notion of a nucleus, which divides any situation into at most three parts: a preparatory phase, a culminating event, and a consequent phase. This nucleus may be manipulated so as to drop or add certain parts, or to interpret its phases in a different way, analogous to Croft’s profiling and conceptual adjustment.

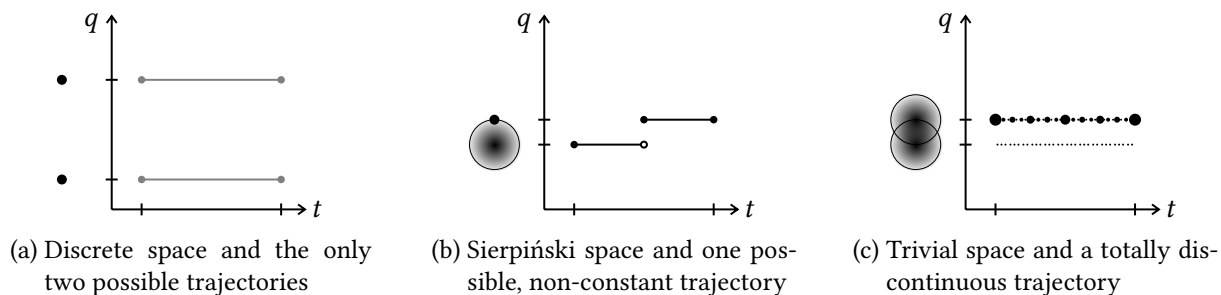


Figure 4: Three two-point spaces and trajectories within them

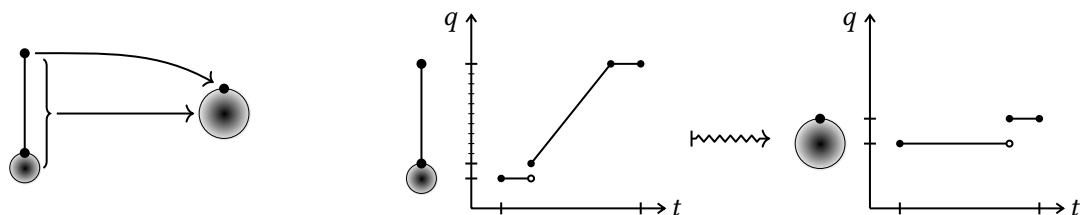


Figure 5: Quotient from an interval to a two-point space, and corresponding effect on the trajectory

any intervening continuum either but does not distinguish the two points anymore, allowing just any transition between them to occur, even totally discontinuous ones (Figure 4c).

Beyond making a collection of points more or less cohesive, distinctions between given points can be forgotten entirely, yielding a truly coarser space via a procedure known as **quotienting**. This is just one type among several transformations that I use to translate the notion of granularity in the qualitative dimension used by Croft (2012); together, these help explain what fine-grainedness and coarse-grainedness might mean at the level of conceptual spaces. Specifically, quotient functions prescribe what distinctions to collapse inside a space, and produce an output space; this operation can therefore relate two spaces together, each associated with its own alternative construal. Thus, a multidimensional, unwieldy conceptual space may be considered under several lenses, each producing a different, more reasonable conceptual space within which to construe the situation. This is exactly what can be done in the passage from Figure 2 to 3, by collapsing all non-result states into one less precise rest state and keeping the result state unchanged: this is represented in Figure 5. This operation suggests that a general-purpose conceptual space is not warranted for the study of aspect; in fact, many construals are even indifferent to the pointwise makeup of the specific conceptual space involved for a certain event type.

In the following, I define aspectual types formally, and formulate additional conditions which exclude many topological spaces and trajectories from consideration. Within the remaining topological spaces, any number of dimensions is allowed in principle, as is assumed for other kinds of conceptual spaces (see Goodale 2022), but only one-dimensional conceptual spaces seem to play a role for aspectual types in practice. I also give a few operations relating different construals together, meant to explain certain predicates' aspectual potential; these operations could constitute an inventory of possible processes inherent to the human conceptual faculty. These proposals provide a first step towards an exhaustive, theoretically motivated and empirically adequate aspectual classification. Because of its newfound precision, the formalism used here provides sharp but flexible tools which can hopefully deepen our understanding of aspect, seen as a manifestation of event conceptualization.

My formalization also has the advantage of making all information precise about each aspectual type and attempts to bring them under one banner, giving a more constrained framework within which to

conjecture and evaluate natural kinds and operations of aspectual construal. For instance, one can define in general an order relation which is total on certain spaces, and could be a crucial element in the definition of directedness. This prompts me to reevaluate the status of nonincremental accomplishments ('runup achievements') and to merge this aspectual type into incremental accomplishments. Because of topological features, this theory does not give substance to distinctions in reversibility, and I ascribe related phenomena to world knowledge and other reasoning mechanisms. Finally, the trajectory required for a cyclic achievement construal seems to be a topological oddity in the system: because of this, I analyze semelfactives as indeterminate between accomplishments and undirected activities, and relate the two aspectual types using a transformation. This partly vindicates an earlier account of semelfactives as activities with naturally atomic minimal events (Rothstein 2008), and provides another view into the coarse-grainedness alternation involved in the aspectual construal of semelfactives.

Section 2 expounds those notions of topological theory that are necessary for a comfortable understanding of the theory. It assumes slightly more than naive set theory, and can safely be skipped by any reader familiar with topology.^d Section 3 explains the mechanisms and interpretation of the theory, as well as its contribution for the study of event conceptualization.

^dIt assumes some comfort with the following notions: sets, functions, relations; propositional and predicate logic; power sets; (pre)orders; natural, rational and real numbers (and intervals thereof); sequences and composition of functions.

2. General topology background

Topology is the study of the internal cohesion of collections of points. These collections are individually called SPACES, and are usually geometric objects. As with most other mathematical structures, topology provides transformations that preserve this cohesion: these are called CONTINUOUS FUNCTIONS between those spaces, and they will be crucial in Section 3. Topology underlies the more widely known $\varepsilon\delta$ -definition of continuity for metric spaces (see Appendix A.3), a special case of topological spaces which also have a notion of distance between their points, compatible with the topology. However, for our purposes we need the more general, flexible formulation provided by general topology, which requires related but distinct intuitions. The goal of this section is to present principles behind general topology, making general (non-metric) topological spaces more accessible to an audience without significant background in mathematics.

2.1. Idea

Two views of continuity The concept of topology emerged relatively late in the history of mathematics, around the turn of the 20th century, to explain the idea of continuity, more specifically that of a continuous function from a source space to a target space.^e Recall that in general, functions are also called maps, mappings, or transformations, and send every element in a specified source set, also called ‘domain’ or ‘input,’ to one element in a specified target set, also called ‘codomain’ or ‘output’; they are then said to go from the former (in)to the latter. There are two ways to conceive of functions from the real line, and relatedly two ways to conceive of their continuousness, one of which is more general than the other.

The physical view Under the physical view, the source is a one-dimensional time interval, and a function is the data of a point for each instant in time. Seen more dynamically, this is a point that can *move across time*, or possibly stay in place. Such a function is then continuous just in case there are no jumps or teleportations of any kind in the trajectory of the point. In other words, there is no need to raise your pen from the paper when tracing out the graph of the function.

The spatial view The more general view is that of a function as a transformation or deformation of the source, so that it fits inside the target. This view can accommodate domains of any shape, not just a time interval. Then, continuity of a function allows all sorts of *bending*, *squeezing* or *gluing* of the shape, but importantly not *tearing* the shape. This can be clarified using the idea of **internal cohesion**, meant to represent the extent to which the points sit together. A merely continuous map between two geometric shapes may discard many aspects of the geometry entirely: distances, angles, even collapsing whole segments or arcs into single points. Indeed, these properties are not what continuity cares about. Instead, continuity preserves the internal cohesion between the points of the shape, keeping into the target space whatever cohesion was there in the first place.

Cohesion These two views are related, in that a line that has been broken into two disconnected segments requires a dynamic point traveling along it to jump when crossing the rupture. In other words, a temporally discontinuous trajectory, if seen as a deformation of the time interval, must involve tearing the time interval spatially in some way. The former view places emphasis on the temporal nature of the domain, while the latter stresses its spatial nature. However, both carry the idea that certain

^eCrucially, not the notion of a “continuous space.” For this, mathematicians talk about *connected* spaces instead (see Definition 2.18). However, both connectedness (of objects) and continuity (of transformations) can only be formulated in the context of topological spaces.

mathematical constructs have some internal cohesion which continuous functions should not tamper with. In keeping with the mathematical tradition, I call these constructs **TOPOLOGICAL SPACES** (‘spaces’ for short), refer to their elements as **POINTS**, and talk about them in spatial terms; this includes the time interval, which is just another example of a space (see Definition 2.3). Functions between spaces that preserve cohesion are called **CONTINUOUS functions** or simply ‘maps.’ Of course, the temporal view of functions will occasionally be useful, when referring to actual events that unfold across time.

2.2. Understanding cohesion through neighborhoods

Neighborhoods are enveloping subsets The pre-formal way to give cohesion to a cohesion-less set X rests on the idea of specifying **neighborhoods** for each of its elements. That is, every point y in a topological space X has a collection $\mathcal{N}_y$ of neighborhoods containing y (subject to a few axioms, see Definition 2.8), which are subsets of X .^f The guiding intuition is that a neighborhood N of a point y is a subset of X which should *surround* y entirely. That is, any such N should encompass every possible route one might take within the space X to “go to y .” Importantly, a neighborhood N is just a collection of points of X with no directionality involved: the idea is simply that one cannot get to y without first crossing N . Reversing perspectives, this is like saying that “there is enough room around $[y]$ in $[N]$ to move in any direction, though perhaps not very far.” (nLab authors 2024) In fact, the whole purpose of neighborhoods is that they can have an arbitrarily small size, so long as they envelop y . The size of a subset can be understood as a relative notion, that of inclusion between sets: a neighborhood N is smaller than another N' if and only if (henceforth iff) $N \subset N'$. In this sense, **the smallest neighborhoods carry information together about what it means to cohere around y .**^g This is cashed out in the next paragraphs.

Key property What does it mean to ‘go to y ’ in the space X , then? The key property to understand the purpose of neighborhoods has to do with the concept of a **limit**. A sequence indexed by the natural numbers $(x_n)_{n \in \mathbb{N}} : \mathbb{N} \rightarrow X$ ^h has y as a **LIMIT** if the sequence gets arbitrarily close to y as n goes to infinity. Other terms for this concept include ‘tend to y ,’ ‘approach y ,’ ‘converge to y .’ The point y is then called ‘a limit’ of the sequence.ⁱ The neighborhoods of a point y specify what it means to get arbitrarily close to y , and therefore to have y as a limit. This is reflected in the following informal but definitional property:

- (5) Getting arbitrarily close to a point y (possibly without reaching it) means that for all neighborhoods N of y , you should reach N *at some point*, and stay within N indefinitely after that.

Crucially, this should be true even of the smallest neighborhoods (in the set-inclusion sense). For instance, let us consider the special case of sequences: say you have a sequence $(x_n)_{n \in \mathbb{N}}$ of points indexed by the natural numbers, and want it to reach the point y *by the time you get to infinity*. Among

^fI often confuse a topological space and its underlying set. Formally, a topological space is really an ordered pair $\langle X, \mathcal{T}_X \rangle$ where $\mathcal{T}_X$ is a topology on X (see Definition A.3). Confusing the two is usually harmless: elements, subsets, etc. pertain to the underlying set, while neighborhoods, opens and continuity are attributes of the topological space.

^gNote that there may not in fact be a smallest neighborhood (a neighborhood included in every other one), or even minimal neighborhoods (neighborhoods that do not include any smaller ones). More precisely, what I mean is that any collection such that any neighborhood is bigger than some member of the collection will carry enough information to understand how to get to y . Such a collection is exactly a neighborhood basis: see Definition 2.2.

^hWhenever X is a set, x is a variable and t is a term referring to a mathematical object (possibly using the variable x), the notation $(t)_{x \in X}$ is just a variant of the functional notation $\lambda x \in X. t$. It is conventionally used for sequences (functions with $\mathbb{N}$ as their domain) and for collections of sets indexed by another set. See e.g. $(X_\alpha)_{\alpha \in A}$ on p. 44 or $(O_\alpha)_{\alpha \in A}$ on p. 66.

ⁱIn some spaces, there can only be at most one limit, so it makes sense to talk about “the” limit of a sequence. This will not be the case for our spaces: see the discussion of the Sierpiński space on p. 16.



Figure 6: The space of natural numbers $\mathbb{N}$

sequences indexed by the naturals, satisfying a property “indefinitely” means ‘for all sufficiently great n ,’ or more precisely ‘there is a threshold n_0 such that the property holds for all $n \geq n_0$.’ Here, the property of interest is “being in the specified neighborhood N ,” or formally, $\lambda x \cdot x \in N$. Making the necessary substitutions into (5), this yields the following definition.

Definition 2.1. Let $(x_n)_{n \in \mathbb{N}} : \mathbb{N} \rightarrow X$ be a sequence of points of X and y be a point of X . Say that y is a **limit** of $(x_n)_{n \in \mathbb{N}}$ as n tends to infinity, and write $(x_n)_{n \in \mathbb{N}} \rightarrow y$, iff:

$$(\forall N \in \mathcal{N}_y) (\exists n_0 \in \mathbb{N}) (\forall n \geq n_0) x_n \in N.$$

Therefore, knowing all neighborhoods of a point y allows us to know what it means to **get arbitrarily close to** y ; in other words, to reach y smoothly, *without any jumps, discontinuities or teleportations*. This is exactly why topology can be used to explain some aspects of the notion of **continuum**.

Neighborhood basis Usually, checking something for a collection of neighborhoods can be tedious. This is why a smaller collection of neighborhoods is usually picked out, such that checking a condition on this collection is enough to prove that fact for all neighborhoods. Such a collection is called a **NEIGHBORHOOD BASIS**, and its elements are called basic neighborhoods. Many different neighborhood bases exist for a given point, but the usual neighborhood bases only have elements of a similar shape – this will be apparent with our very first examples of topological spaces in Section 2.4.

Definition 2.2. Let X be a topological space, with a collection of neighborhoods being given for each point: $(\mathcal{N}_x)_{x \in X}$. Let x be a point and $\mathcal{B}$ be a subset of $\mathcal{N}_x$, i.e. a collection of neighborhoods of x . The subset $\mathcal{B}$ is a **neighborhood basis for** x iff every neighborhood of $\mathcal{N}_x$ contains some element of $\mathcal{B}$:

$$(\forall N \in \mathcal{N}_x) (\exists B \in \mathcal{B}) B \subseteq N.$$

If not for the special assumptions about the neighborhood basis, there would be no way of ensuring that something which is only true on some subcollection of neighborhoods is also true for the whole set $\mathcal{N}_x$. However, conditions on neighborhoods are usually monotonous: they are predicates P such that if $N \subseteq N'$ and $P(N)$ then $P(N')$. This kind of predicate is more restrictive for smaller subsets, so that anything true on a basis will be true of the whole set. Consider the definition of a limit: assume every basic neighborhood is reached indefinitely by a sequence. Then for any neighborhood N there is always a basic neighborhood, say B , which is smaller ($B \subseteq N$), and is known to be reached indefinitely; but since belonging to B is stronger than belonging to N , then automatically N will be reached indefinitely, too. This shows that checking a limit just on a neighborhood basis of the candidate limit point is enough to prove that it is indeed a limit. More generally, in case we have a neighborhood basis, any property which requires universal quantification of a monotonous predicate over neighborhoods may be checked just on the basis. This hugely facilitates reasoning about topological statements and properties.

2.3. Caveats

Neighborhood of a point Some possible misconceptions should be dispelled right away. First, the idea of being a neighborhood is always relativized to a specific point; there is no “being a neighborhood” *tout court*. Unlike some uses of the word in everyday language, a mathematical neighborhood is not just “a particular region,” but any region *around* some specific point. Relatedly, neighborhoods are a primitive notion that applies to subsets of X , without any underlying, symmetric notion of ‘neighbor’ that would hold of points of X . In particular, there is no conventional notion of what it means for two points x, y to be “neighbors.”^j

Difference with the concept of a distance Second, terms like ‘closeness’ and ‘far away’ should not suggest any assumption of distance within the space. Rather, they are always relative, and reflect set-theoretic inclusion between neighborhoods: a sequence that tends to a point y ends up getting closer to that point because its values are elements of successively smaller neighborhoods of y . Despite this figure of speech, topology abstracts away from notions of distance by design. These two notions are related, though: when a set is equipped with a notion of distance, a natural topology is defined using that distance (see Definition A.11). But topology is more general: it actually makes sense in topology to have *distinct* points x and y that nevertheless sit “at distance 0” from one another, loosely speaking. This can happen if any neighborhood of x contains y , for instance: then, x is so close to y that a constant sequence at y can still be said to tend to x in the sense of Definition 2.1: see p. 16.

Symmetry Furthermore, note that while a distance function is usually required to be symmetric, no such symmetry requirement holds in topological spaces, despite the symmetry of the “neighbor” relation in everyday language. Indeed, the collections of neighborhoods of two nearby points x, y might behave very differently from one another. As before, the two points might be such that every neighborhood of x contains y , while y has as one of its neighborhoods the singleton set $\{y\}$, which does not contain x .^k This is an especially interesting idea, which allows a regimented notion of closeness without declaring the two points indistinguishable (which would constitute a trivial kind of topology); see p. 16.

Order and direction Although any neighborhood should *intuitively* capture every direction around a point, this is again metaphorical and does not presuppose any notion of order or direction within the space. This sets topological spaces apart from linearly ordered sets, which have a direction given by the order relation, and from vector spaces, which have global directions given by lines together with the additive structure. Again, both of the constructs can interact meaningfully with topology, but the latter is more general.

2.4. Some examples

Let us consider two examples of topological spaces, by describing what their neighborhoods look like.

^jA possible notion could be “ x is a neighbor of y iff x is a member of all neighborhoods of y , i.e. $(\forall N \in \mathcal{N}_y) x \in N$ ” but as we will see on p. 16 this is not a symmetrical notion. Other, symmetrical candidates for this notion are usually given different names (e.g. “ x is in all neighborhoods of y and vice-versa” is usually called **indistinguishability**: see Appendix A.1).

^kA symmetry requirement can be enforced, gives rise to the seldom used notion of *symmetric* topological space. This is a quite weak notion that is usually less useful than the stronger T_1 separation axiom; for a definition of both of these, see Appendix A.1.

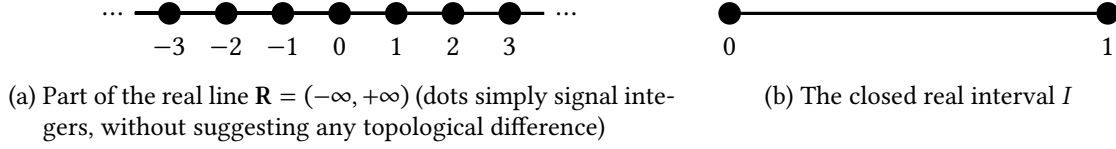


Figure 7

The real interval One simple space of interest to us is the closed real interval $[0, 1]$, which I also notate I , and define as the set of real numbers (see Figure 7a) between 0 and 1, inclusive: see Figure 7b. Recall that each real number has a potentially infinite decimal expansion: numbers in this interval are those real numbers whose integer part is 0.

Definition 2.3. The closed real interval, notated with $[0, 1]$ or I , is defined as:

$$[0, 1] := \{y \in \mathbf{R} \mid 0 \leq y \leq 1\}.$$

All points apart from the two endpoints, such as the midway point $1/2$, cohere to the space in the same way, with a continuum on either side. Any small part of the continuum which sticks to or kisses $1/2$ on either side should then count as “surrounding” $1/2$, i.e. being a neighborhood thereof. This is exactly what the formal definition of the topology on $[0, 1]$ states: the neighborhoods of $1/2$ are exactly those subsets of $[0, 1]$ which contain some (possibly very small) interval $(1/2 - \varepsilon, 1/2 + \varepsilon)$ around $1/2$; any such interval indeed surrounds the point $1/2$, provided $\varepsilon > 0$.

Neighborhoods are of the exact same kind for any point of the interval that is not 0 or 1. This is fitting because, as the endpoints of the interval, the points 0 and 1 cohere to the space in a different way. Indeed, surrounding 0 *within the space* $[0, 1]$ can only be done from the right: there is simply nothing on the left of 0 within that space. Thus, the neighborhoods of 0 are the subsets which contain some right-facing interval $[0, 0 + \varepsilon)$ containing 0 (with $\varepsilon > 0$). The point 1 is entirely analogous, swapping right and left.

Definition 2.4. The neighborhoods in the closed real interval I are defined in the following way:

$$\begin{aligned} \text{if } y \in (0, 1): & \quad \mathcal{N}_y := \{N \subseteq I \mid (\exists \varepsilon > 0) (y - \varepsilon, y + \varepsilon) \subseteq N\}, \\ \text{if } y = 0: & \quad \mathcal{N}_0 := \{N \subseteq I \mid (\exists \varepsilon > 0) [0, \varepsilon) \subseteq N\}, \\ \text{if } y = 1: & \quad \mathcal{N}_1 := \{N \subseteq I \mid (\exists \varepsilon > 0) (1 - \varepsilon, 1] \subseteq N\}. \end{aligned}$$

Here, we see our first example of a neighborhood basis: for any point $y \in (0, 1)$, the collection $\{(y - \varepsilon, y + \varepsilon) : \varepsilon > 0\}$ defines a neighborhood basis; that is, a subset of $\mathcal{N}_y$ such that any neighborhood is a superset of some basic neighborhood. Many other neighborhood bases can be defined, such as those where the interval is not symmetric around y ; in fact, the collection of all open intervals containing y is another example of a neighborhood basis. However, it is just as simple to single out the symmetrical intervals, because any unbalanced interval contains some symmetrical interval and vice-versa.

Thanks to this remark, we may simply leave out the bigger, less restrictive neighborhoods in the definition of a limit and arrive at the entirely equivalent, simpler statement (6).

$$(6) \quad (x_n)_{n \in \mathbb{N}} \rightarrow y \iff (\forall \varepsilon > 0) (\exists n_0 \in \mathbb{N}) (\forall n \geq n_0) x_n \in (y - \varepsilon, y + \varepsilon).$$

Note that the intuition of the interval, where a neighborhood basis consists only of symmetrical areas around the point, transposes in an entirely analogous way to neighborhoods of any metric space (including higher-dimensional spaces, see Section 2.10). This happens simply by replacing open intervals around a point by open balls, sets of points which lie less than a fixed distance away from the initial point.

The Sierpiński space To get a glimpse of the generality of topological spaces as opposed to metric spaces, let us consider a non-metrizable topological space (see Definition A.12). One case that does not arise in metric spaces is when a point y is found in every neighborhood of another point x . This is because the notion of distance automatically separates any two distinct points with respective neighborhoods that are also disjoint.^l Counterexamples to this principle bring us into the realm of so-called **non-separated topologies**.^m Consider the space S with exactly two points, g and c , and declare both subsets that contain g (in particular, $\{g\}$ itself) to be neighborhoods of g , while the only neighborhood of c is the entire space.

Definition 2.5. Define the **Sierpiński space** as the two-point set $S := \{g, c\}$, with respective collections of neighborhoods defined in the following way:

$$\begin{aligned} \text{if } y = g: & \quad \mathcal{N}_g := \{\{g\}, \{g, c\}\} = \{N \subseteq S \mid g \in N\}, \\ \text{if } y = c: & \quad \mathcal{N}_c := \{\{g, c\}\} = \{S\}. \end{aligned}$$

In that case, g indeed is found in every neighborhood of c , because there is only one. This is the asymmetric situation evoked on p. 14. To understand what this means, consider which sequences in the space converge to either point. As was already pointed out, the constant sequence at g converges to c ; in fact, any arbitrary sequence will converge to c , because it stays indefinitely within the only neighborhood of c . On the other hand, per Definition 2.5, a neighborhood basis of g is given by the subset of $\mathcal{N}_g$ whose only member is the smallest neighborhood of g , the singleton $\{g\}$; this neighborhood basis is the set $\{\{g\}\}$. Because this is a neighborhood basis, the only way to get arbitrarily close to g is to actually reach and stay within $\{g\}$ “before time is up,” and this means the only sequences for which g is a limit are those that are constant at g after some threshold n_0 (these sequences are called ‘almost constant’). This is because in a sense, g surrounds *itself*. In other words, there is no sequence of points that *approaches* g , strictly speaking. Note that an almost constant sequence at g also has c as a limit: the notion of a limit is not unique in general topological spaces.ⁿ As a summary, there are no restrictions on getting to c , but there is a heavy restriction on reaching g ; and this even though both points sit “adjacent” to each other!

One line of explanation for this counterintuitive fact appeals to the idea of closed properties. A closed property is a property of points of a space, that is *preserved when the limit of a sequence is taken*, or “passes to the limit” in mathematical jargon. (Note that the definite article is an abuse of language that should not suggest that limits are unique.)

^lSee Section A.3 and Proposition A.13.

^mSee Axiom T₁ in Appendix A.1.

ⁿIt is unique in T₁ spaces, however, and a fortiori in metric spaces: see Appendices A.1 and A.3.



Figure 8: The Sierpiński space S , with the open point g shown with a fading interior.

Definition 2.6. Let P be a property of points of X and $(x_n)_{n \in \mathbb{N}}$ be a sequence of points in X . The property P is **closed** (or **passes to the limit**) iff, whenever y is a limit point of (x_n) and P holds indefinitely (i.e. near infinity) on the sequence (x_n) , then $P(y)$ holds:

$$(\forall y \in X) [(x_n) \rightarrow y \wedge (\exists n_0 \in \mathbb{N}) (\forall n \geq n_0) P(x_n)] \implies P(y).$$

A sequence of points may gain such a property as it passes to the limit, but it may not lose the property if it has it “near infinity” in the first place. This idea explains the asymmetric convergence of sequences in this space if we assume for the sake of the argument that c has some closed property that distinguishes it and makes it more “specific” than g . A limit of a sequence that visits c every so often should indeed still satisfy that property; so here the limit should be c itself. By contrast, the only way for the sequence to never be “contaminated” by the special property is to simply stay indefinitely at g after a while. On the other hand, a closed property may be gained at the end of time even though you never had it in the first place. This is why the only sequence that tends to g is a sequence that eventually ends up at g , while there is nothing special to do to reach c . In that case, g is said to be an **OPEN POINT**. In my diagrams, **open points that sit adjacent to other points are shown with a fading interior**, inspired by the representation in Eisenbud and Harris (2000). For instance, the Sierpiński space will be given the representation in Figure 8.

Definition 2.7. Let x be a point of X . The point x is **open** iff $\{x\}$ is a neighborhood of x , or equivalently^o if any subset that contains x is a neighborhood of x :

$$\mathcal{N}_x = \{N \subseteq X \mid x \in N\}.$$

Discrete space Notice that in the previous example, the fact that the point g was open forced any converging sequence to g to actually end up at g . In other words, g already surrounds itself, and the only way to get to it is to reach it. In the case of a space where all points are open, we end up with a so-called **discrete space**. This is a kind of space that actually has essentially no cohesion between its points, so that (a) continuous maps from it into any topological space are not more constrained than any usual set map; (b) any function into it should collapse into a single point all components with some internal cohesion (called ‘connected components,’ see Section 2.9). The space of natural numbers $\mathbb{N}$, depicted in Figure 6, is an example of a discrete space.

Trivial spaces The opposite situation is when there are as few neighborhoods as possible: this happens when the only neighborhood of any point is the whole space itself. Then, all points belong to the only neighborhood of every other point, so that the topology does not make any difference between the points: they all cohere together in the tightest blob. Every sequence converges to every

^oThe equivalence follows from Axiom N3 to be seen shortly.

point since it stays indefinitely within all neighborhoods of every point. More generally, continuous functions into trivial spaces do not have any restrictions; conversely, continuous functions from trivial spaces can only send the whole space to one point, or else to mutually indistinguishable points (see Definition A.1). An example of a trivial space was given in Figure 4c on p. 9, along with a totally “discontinuous” function into it (for instance, sending all rational numbers to one point and irrationals to another), which is still a legal continuous function.

2.5. Definition of a topological space

To make this exposition complete, I present the axioms on neighborhoods that are required to hold no matter the space. Note that this still leaves a lot of room for variation within the class of topological spaces, as we have just seen. I leave to the reader to check that all the previous examples of systems of neighborhoods satisfy these properties.

Definition 2.8. Let X be a set. The data $(\mathcal{N}_x)_{x \in X}$ of a collection of neighborhoods for each point should satisfy the following points:

Axiom N1. *The entire space is a neighborhood of x :*

$$(\forall x \in X) X \in \mathcal{N}_x.$$

Axiom N2. *Any neighborhood of x contains x :*

$$(\forall x \in X) (\forall N \in \mathcal{N}_x) x \in N.$$

Axiom N3. *Any superset of a neighborhood of x is still a neighborhood of x :*

$$(\forall x \in X) (\forall N \in \mathcal{N}_x) (\forall N' \in \mathcal{P}_X) (N \subseteq N' \implies N' \in \mathcal{N}_x).$$

Axiom N4. *The intersection of any two neighborhoods of x is still a neighborhood of x :*

$$(\forall x \in X) (\forall N, N' \in \mathcal{N}_x) N \cap N' \in \mathcal{N}_x.$$

This last axiom is slightly more abstract, and applies to all neighborhood data together, not just each $\mathcal{N}_x$ individually.

Axiom N5. *If N is a neighborhood of x , then the set of points that N is a neighborhood of (called the ‘interior’ of N) is still a neighborhood of x :*

$$(\forall x \in X) (\forall N \in \mathcal{N}_x) \{y \in X \mid N \in \mathcal{N}_y\} \in \mathcal{N}_x.$$

The first four axioms all vindicate the intuition that neighborhoods should surround a given point. In order: (N1) The whole space does indeed surround the point. (N2) Any area that surrounds a point contains the point: this is useful because staying at x should also count as converging to x . (N3) A superset of an area that surrounds a point still surrounds that point. (N4) If two areas both surround a point, the area where they overlap also surrounds the point.

Axiom N5 is harder to interpret, but it basically enforces that points do not have incoherent neighborhoods. Indeed, it places a condition on neighborhoods of other points than x : any neighborhood of x can be stripped of its outer elements (those of which it is not a neighborhood) but still remain a neighborhood of x . In other words, “having N as a neighborhood” ($\lambda x \cdot N \in \mathcal{N}_x$) is a property that can

only hold of points that sit comfortably inside N (that is, surrounded by points which also have N as a neighborhood).^P This axiom would be violated in the following three-point space:

Definition 2.9. Let Z be the three-pointed set $\{a, b, c\}$. Define a collection of neighborhoods for each point by:

$$\begin{aligned}\mathcal{N}_a &:= \{N \subseteq Z \mid a \in N\}, \\ \mathcal{N}_b &:= \{N \subseteq Z \mid b \in N \wedge a \in N\}, \\ \mathcal{N}_c &:= \{N \subseteq Z \mid c \in N \wedge b \in N\}.\end{aligned}$$

Then, consider the neighborhood $N_0 := \{b, c\}$ of c . It is not a neighborhood of all points sufficiently close to c , because N_0 does not appropriately *surround* b (since it does not also contain a). But then, the set $\{y \in Z \mid N_0 \in \mathcal{N}_y\}$ is the singleton $\{c\}$, and this alone does not count as a set of points “sufficiently close to c ” (as b is infinitesimally close to c). In particular N_0 does not surround any neighborhood of c . If we remove this neighborhood instead, then we get $\mathcal{N}_c = \{Z\}$ so that now all neighborhoods of c comfortably surround c . Among other things, this forces the relation $\lambda xy \cdot (\forall N \in \mathcal{N}_x) y \in N$ ^Q to be transitive: therefore, if the constant sequence at a point y converges to x , and the constant sequence at x converges to z , then the point z is also a limit of the constant sequence at y .

Together, these five axioms on the collections of neighborhoods $(\mathcal{N}_x)_{x \in X}$ define a notion of **TOPOLOGICAL SPACE**.

Definition 2.10. A **topological space** is a set X together with the data, for all $x \in X$, of a collection of neighborhoods $\mathcal{N}_x \subseteq \mathcal{P}_X$, satisfying Axioms N1–N5 above.

This is one of several equivalent ways to define a topological space, through which all topological notions can be defined. A more abstract and condensed formulation is more common, which I touch upon in Section 2.8 and Appendix A.2. The current formulation has the advantage of being more easily interpretable in terms of cohesion.

2.6. General continuous functions

Taking stock, we have defined neighborhoods and sketched how they can be used to reflect the ideas of cohesion and convergence. This was enough to define topological spaces, but we have yet to define a general notion of continuity (preserving cohesion) for these new objects. I will first explain how converging sequences are a special case of continuous functions, and analyze this case to distill a definition of continuity.

The space of natural numbers The only real use of neighborhoods we have seen up until now was for sequence convergence, through Definition 2.1. This restriction was initially justified though, because looking at all sequences converging to a point together is very often enough to understand how the space coheres around a given point. This usefulness of sequences is reflected by the “sequential characterization of limits and continuity,” a property which holds in most usual topological spaces,

^PBourbaki (1971, TG I.3) expresses this as follows: “any neighborhood of x is also a neighborhood of all points sufficiently near to x ” (where ‘sufficiently near to x ’ means ‘for some neighborhood of x ’).

^QOr $\lambda xy \cdot x \in \overline{\{y\}}$, see Definition 2.17.

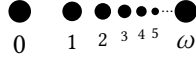


Figure 9: The space of natural numbers N_ω with a limit point ω

including the ones considered here.^r In fact, Definition 2.1 is not far from a much more general definition of continuity, which should of course allow many source spaces other than N . Indeed, a sequence $(x_n)_{n \in N}$ in X that converges to y can actually be interpreted as a continuous function *from the space* $N_\omega := N \cup \{\omega\}$ into X , where ω is the “point at infinity,” or “infinity” for short. The space N_ω is shown in Figure 9; its topology will be defined shortly (Definition 2.11). This corresponding continuous function

$N_\omega \rightarrow X$ is defined by $\lambda n \in N_\omega \cdot \begin{cases} x_n & \text{if } n \in N \\ y & \text{if } n = \omega \end{cases}$. Note that ω , which is the limit of the naturals $(n)_{n \in N}$ in

N_ω , is sent to a limit of the sequence $(x_n)_{n \in N}$ in X , namely y . To see that this is a continuous function, first notice that because each point in N is open, there are no restrictions on the image of each natural number as I announced on p. 17. However, the point ω is not an open point, because natural numbers end up “accumulating” near it. Thus, continuity forces a condition on its image by the function, which is dictated by the limiting behavior of the function on the natural numbers: the sequence $(x_n)_{n \in N}$ must converge to the image of ω , otherwise the cohesion of the space would be broken when going through the function.

Sequence convergence as continuity Let us go back to the informal key property behind neighborhoods, in (5) on p. 12. It was phrased in a temporal fashion reminiscent of sequences, but we can rephrase it in more spatial terms. There are actually two limiting points involved in a converging sequence: the limit in the source space, the point ω , and the limit in the target space, y . Thus, a sequence getting closer to a limit point actually mimics in the target space X the “canonical” way to get closer to a specific point, namely that found in the source space N_ω as N transitions into ω . The key insight is that this mimicking behavior of the converging sequence is simply one avatar of continuity, since it involves *preserving the cohesion of the source space* into the target space. Under this view, Definition 2.1 exactly expresses the “continuity at the point ω of the domain,” as announced in the previous paragraph. We can now rephrase this property of continuity in terms of **neighborhoods of ω** , which will ease the transition into the more general notion of continuity.

Neighborhoods of ω Indeed, a neighborhood basis for ω in the space N_ω is given by all discrete intervals of the shape $\llbracket n_0, \omega \rrbracket := \{n \in N \mid n \geq n_0\} \cup \{\omega\}$, or more concisely $\{n \in N_\omega \mid n \geq n_0\}$,^s with one such interval for each natural number $n_0 \in N$. Let us call this neighborhood basis $\mathcal{B}_\omega := \{\llbracket n_0, \omega \rrbracket : n_0 \in N\}$. Together with the fact that N only has open points, we can now define the topology of N_ω .

Definition 2.11. Let N_ω be the set $N \cup \{\omega\}$ where ω is not a natural number. Define a topology on N_ω with the following neighborhoods:

$$\begin{aligned} \text{if } n \in N: & \quad \mathcal{N}_n := \{N \subseteq N_\omega \mid n \in N\}, \\ \text{if } n = \omega: & \quad \mathcal{N}_\omega := \{N \subseteq N_\omega \mid (\exists B \in \mathcal{B}_\omega) B \subseteq N\} \\ & = \{N \subseteq N_\omega \mid (\exists n_0 \in N) \llbracket n_0, \omega \rrbracket \subseteq N\} \\ & = \{N \subseteq N_\omega \mid (\exists n_0 \in N) (\forall n \in N_\omega) n \geq n_0 \implies n \in N\}. \end{aligned}$$

^rIt holds in all *sequential spaces*, a very general class of spaces including all metric spaces, all countable spaces, and more.

^sThis involves making N_ω into a poset by adding to the order in N the ordering $n \leq \omega$ for all $n \in N$.

To get arbitrarily close to infinity within the space $\mathbf{N}_\omega$, you should reach any neighborhood of ω indefinitely; in other words, you should surpass any n_0 at some point and never go back below it. A very important realization is that, for a sequence converging into some arbitrary space X , **the set of points that reach any given neighborhood of y must itself be a neighborhood of ω** . To see this, let us reformulate Definition 2.1:

$$\begin{aligned} (\forall N \in \mathcal{N}_y) (\exists n_0 \in \mathbf{N}) (\forall n \geq n_0) x_n \in N &\iff (\forall N \in \mathcal{N}_y) (\exists n_0 \in \mathbf{N}) (\forall n \in \mathbf{N}_\omega) n \geq n_0 \implies x_n \in N \\ &\iff (\forall N \in \mathcal{N}_y) (\exists n_0 \in \mathbf{N}) \llbracket n_0, \omega \rrbracket \subseteq \{n \in \mathbf{N}_\omega \mid x_n \in N\} \\ &\iff (\forall N \in \mathcal{N}_y) (\exists B \in \mathcal{B}_\omega) B \subseteq \{n \in \mathbf{N}_\omega \mid x_n \in N\}. \end{aligned}$$

This means that the set $\{n \in \mathbf{N}_\omega \mid x_n \in N\}$ on the right, which does not depend on n_0 , contains some basic neighborhood of ω . But recall from Definition 2.2 that containing a basic neighborhood of a point is exactly the same as being a neighborhood of that point. Therefore, the above property rewrites as:

$$(7) \quad (\forall N \in \mathcal{N}_y) \{n \in \mathbf{N}_\omega \mid x_n \in N\} \in \mathcal{N}_\omega.$$

Continuity in terms of preimages This concept of “the set of points that are sent by the function a given subset of the codomain,” as exemplified by the set $\{n \in \mathbf{N}_\omega \mid x_n \in N\}$ above, is so useful in general that it has a special name: the inverse image, or **PREIMAGE** of a target subset.

Definition 2.12. Let $f : X \rightarrow Y$ be a set map, and B be any subset of Y . The **preimage of B by f** , notated $f^*(B)$, is defined as the set of points of X that are sent inside B by f :

$$f^*(B) := \{x' \in X \mid f(x') \in B\}.$$

In the case where $X = \mathbf{N}_\omega$ and f is a sequence so that $f(n)$ is notated x_n , this exactly matches the neighborhood of infinity in (7). Having a concept for this set allows us to have a very concise definition of continuity at a point:

Definition 2.13. Let X, Y be two topological spaces, $f : X \rightarrow Y$ be a function between their underlying sets, and x be a point of the domain X . Say that f is **continuous at x** iff the preimage of any neighborhood of $f(x)$ in Y is a neighborhood of x in X :

$$(\forall N \in \mathcal{N}_{f(x)}) f^*(N) \in \mathcal{N}_x,$$

or equivalently:

$$(\forall N \in \mathcal{N}_{f(x)}) \{x' \in X \mid f(x') \in N\} \in \mathcal{N}_x.$$

Once we universally quantify over the point x above, we can say that f is **continuous** without any qualification:

$$f \text{ is continuous} \stackrel{\text{def}}{\iff} (\forall x \in X) (\forall N \in \mathcal{N}_{f(x)}) f^*(N) \in \mathcal{N}_x.$$

That is, in looser prose: whatever the level of closeness required around $f(x)$, the collection of points that come at least this close to $f(x)$ surrounds x . This notion of “surrounding” is primitive in general topological spaces, being prescribed by the neighborhoods. In the case of a metric space, it is expressed in terms of closeness: in that setting, for a collection to “surround x ” means that there is a real number, another threshold of closeness below which all points within that distance to x belong to the collection.

We can now make clear the intellectual progression between this definition of continuity at a point and the classical $\varepsilon\delta$ -definition of continuity for metric spaces. Semi-formally, this latter definition states that a function between metric spaces is continuous at a point x iff, for any fixed distance threshold ε in the output, there is a threshold δ in the input such that all points that are δ -close to x are sent ε -close to $f(x)$. The correspondence between the two is obvious, and they do in fact exactly coincide, if both metric spaces are given the canonical topology (see Definition A.11 and Proposition A.14).

Preserving cohesion or respecting separation I started this exposition by presenting continuity as *preserving cohesion of the source space into the target space*. The final definition actually reverses this notion somewhat, because it is neighborhoods in the target space whose preimages must retain neighborhoodness in the source space. One insightful way to make sense of this could be the following remark: even though together, neighborhoods reflect cohesion in the space, each of them in fact embodies some notion of **separation**. A neighborhood of x actually separates the point x within from the points without. The fewer neighborhoods there are, the more “cohesive” the space is (culminating with the trivial topology); the more open sets there are, the more “separated” the space is (culminating with the discrete topology). Therefore, the preservation of neighborhoods by preimage expresses that if two points are separated in the image, then they should also have been separated in the input. In other words, the function should not add any separation between the points as it sends them into the target space. The shift from *preserving cohesion* to *not adding dispersion or separation* is also motivated by the following remark: neighborhoods reflect “surroundingness” in the input, but a map can be continuous at x without surrounding $f(x)$ in the target space; hence the forward image of a neighborhood^t being a neighborhood is not a good criterion. However, a continuous map should never introduce any separation among the points, and this is checked by ensuring that, for any separating subset in the output, its corresponding subset in the input is separating: the preimage of a neighborhood is a neighborhood.

Equivalent definitions We have now reached a quite streamlined definition of continuity at a point (Definition 2.13), phrased in a general way that does not require any notion of distance, sequentiality or otherwise. It is applicable to any topological space as we have defined them (in terms of neighborhoods; Definition 2.10), and the unqualified notion of continuity simply follows by universal quantification. One remaining feature of it is that it is couched in local, pointwise terms. This is not necessary, as there exists another, global formulation, that does not quantify over points. This formulation uses open subsets, which I will present briefly; for the definition of continuity in terms of open subsets, I refer the reader to Section A.2. Before that though, I give a taste of what topology might be useful for in the study of event conceptualization, as a preliminary for Section 3.

2.7. Important examples of continuous maps

While continuous maps are easily understood on simple spaces such as the interval, more exotic spaces like the Sierpiński space are not as straightforward. Getting ahead on Section 3, I give several examples of continuous functions $I \rightarrow S$, to show how they can represent the idea of instantaneous transition.

^tGiven a function of sets $f : X \rightarrow Y$ and a subset A of X , the forward image of A is simply the set of images of elements of A : $f_*(A) := \{f(x) : x \in A\} \subseteq Y$.

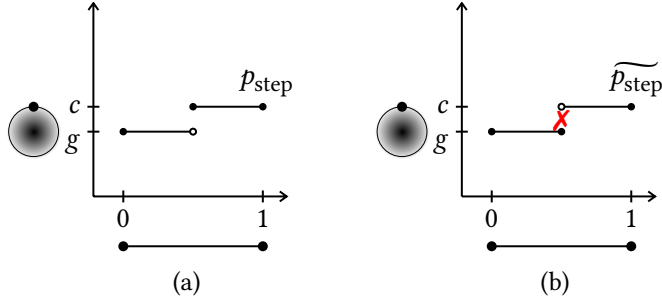


Figure 10: A continuous function $I \rightarrow S$ and a discontinuous one

Consider the function $p_{\text{step}} : I \rightarrow S, t \mapsto \begin{cases} g & \text{if } t < 1/2 \\ c & \text{if } t \geq 1/2 \end{cases}$, represented in Figure 10a. We can check that it is continuous in the following way: for any point $t \in [1/2, 1]$, we have $p_{\text{step}}(t) = c$, and should check that the preimages of all neighborhoods of c are neighborhoods of t . But the only neighborhood of c is the whole space S , and $p_{\text{step}}^*(S)$ is just I which is automatically a neighborhood of t . Now, for any point $t \in [0, 1/2)$, we have $p_{\text{step}}(t) = g$, and the point g has two neighborhoods. For the total neighborhood S , this does not impose any condition on p_{step} , as above with c . However, because $\{g\}$ is a neighborhood of g , we know that $p_{\text{step}}^*(\{g\})$ should be a neighborhood of all $t \in [0, 1/2)$. In other words, all points surrounding any $t \in [0, 1/2)$ should also be sent to g . This is true, because the set of points that are sent to g is $[0, 1/2)$, which is a neighborhood of all its points. Finally, we conclude that p_{step} **is a continuous function**.

Note that for any transition into c , there is no *final moment* at which the trajectory is at g : indeed, continuity forces some immediately subsequent moments to also be sent to g . This still allows a transition to happen, but only *at the first point sent to c* , not before. By the same principle, any transition back to g does not have an *initial moment* which is sent to g . With the reasoning above, we deduce a general condition on functions into S .

Proposition 2.14. Let X be a topological space and $f : X \rightarrow S$ be a function of sets. The function f is continuous iff the preimage of $\{g\}$ by f is a neighborhood of all its points:

$$(\forall x \in X) x \in f^*(\{g\}) \implies f^*(\{g\}) \in \mathcal{N}_x.$$

This characterization shows that the very similar function $\widetilde{p}_{\text{step}} : I \rightarrow S, t \mapsto \begin{cases} g & \text{if } t \leq 1/2 \\ c & \text{if } t < 1/2 \end{cases}$ is not in fact continuous at the point $1/2$. Indeed, $\widetilde{p}_{\text{step}}^*(\{g\})$ is equal to $[0, 1/2]$ which is not a neighborhood of $1/2$; but this fact should be enforced by continuity. Together, the two functions p_{step} and $\widetilde{p}_{\text{step}}$ show that topology, in the interaction between the Sierpiński space S and the closed real interval I , can describe a **regimented notion of instantaneous transition between two adjacent points**. This will of course be useful in representing achievements.

The same is true for transformations of the Sierpiński space; let $\varphi : S \rightarrow S$ be a continuous function. By Proposition 2.14, it must only send to g a set of points which is a neighborhood of all its points. This leaves three choices:

- $\varphi = \text{const}_c : x \in S \mapsto c$, which sends both points to c ;
- $\varphi = \text{const}_g : x \in S \mapsto g$, which sends both points to g ;

- $\varphi = \text{id}_S : x \in S \mapsto x$, which sends either point to itself.

In particular, it is not possible to exchange the two points g and c ; if c is to be sent to g , then φ must also collapse the distinction between c and g . This provides a **flexible but not overly expressive method of transforming spaces**, which will be used in Section 3.6.

2.8. Open subsets

Up to now, specifying a topology on a set amounted to listing a primary collection of neighborhoods for each and every point of x , which are distinguished subsets subject to a few axioms. It turns out that neighborhoods can be defined as secondary objects instead, when one takes OPEN SUBSETS as the primary piece of data (see Definition A.4). Open subsets are also distinguished subsets with certain conditions, but they are crucially not tied down to any point; they are, in a sense, global. The two pieces of data are perfectly equivalent, however: open subsets can also be defined from neighborhoods as secondary objects. Here, I take neighborhoods as primary.

Definition 2.15. Let X be a topological space in the sense of Definition 2.10, with $(\mathcal{N}_x)_{x \in X}$ as neighborhoods. Let O be a subset of X . The subset O is **open** iff it is a neighborhood of every point within it:

$$(\forall x \in X) x \in O \implies O \in \mathcal{N}_x.$$

The collection of all open subsets is written $\mathcal{T}_X$ and called **the topology** of the space X .

Equivalence The topology always satisfies certain conditions (whether it is a primary or secondary object), which are found in Definition A.3. It can be shown that, if neighborhoods satisfy Axioms N1–N5, they can be redefined via the open subsets to yield the exact same neighborhoods as one started with. Conversely, if one starts with a topology satisfying the aforementioned conditions, one can redefine neighborhoods and opens subsets and gain the same topology back. This fact shows that the two definitions are equivalent.

Simplification Usually though, a topology is a simpler piece of data than listing neighborhoods for each point. It can be simplified further by giving only a subset of the topology, a ‘basis’ whose elements are called ‘basic opens,’ which is then sufficient to define the topology as a whole; in just the same way as a neighborhood basis for a point is enough to define all neighborhoods of that point. To see how giving a topology can be simpler than listing neighborhoods, let us consider three examples.

In the Sierpiński space S , the point g is open: any subset containing g is a neighborhood of g . On the other hand, only the whole space S is a neighborhood of c . In terms of opens, it is enough to give the list of open subsets: $\{\emptyset, \{g\}, \{g, c\}\}$. In fact, the set $\{\{g\}, \{g, c\}\}$ is a basis for the topology. Note that the reason why I call g an ‘open point’ is precisely because the singleton $\{g\}$ is an open subset of the space S .

Meanwhile, in the space of natural numbers $\mathbb{N}$ which is discrete, the collection of neighborhoods is one of the simplest possible: all subsets that contain a point n are neighborhoods of it. For open subsets, this is true without the dependency on n : any subset is an open subset. In the case of the slightly bigger and more interesting space $\mathbb{N}_\omega$, open subsets are of two forms instead: either they do not contain ω , or they contain a whole neighborhood of ω , i.e. an interval of the form $\llbracket n_0, \omega \rrbracket$.

Finally, on the real line $\mathbb{R}$, open subsets themselves do not enjoy a simpler description than what follows from Definition 2.15. However, a very simple basis is available: open intervals are a basis of the topology on the real line. On the closed interval, a basis is given by restricting general open intervals



Figure 11

to the subset I . This is done by intersecting (a, b) (if $a, b \in \mathbf{R}$) with $I = [0, 1]$, which essentially gives three types of basic opens:

- open intervals (a, b) with $a, b \in [0, 1]$;
- half-closed intervals $[0, b)$ with $b \in (0, 1]$ (if $a < 0$);
- half-closed intervals $(a, 1]$ with $a \in [0, 1)$ (if $b > 1$).

Note that the only half-open intervals which are actually open subsets in I are those given above, which contain either 0 or 1. While this description involves listing cases, this is bound to happen in a space which is not homogeneous, and this was the case in Definition 2.4, too. However, the description length is slightly smaller here.

Finally, if one removes the end points from the closed interval $[0, 1]$, yielding the open interval $(0, 1)$, the open intervals (a, b) within $(0, 1)$ do actually provide a legitimate base of opens, which enjoys a simpler description than in the case of the closed interval $[0, 1]$, those that are really open intervals. Actually, this open interval is a topological space which is **homeomorphic** to the real line $\mathbf{R}$; that is, isomorphic in the sense of topological spaces. Indeed, intuitively both are linearly ordered, “dense,”^u and have no endpoints: see Figure 11. Because distance in the strict sense is not a concern for topology, this allows us to show that these two spaces are equivalent: they have exactly equivalent points and cohesion, and thus behave in indistinguishable ways, from the point of view of topology. However, presence of endpoints does very much matter for topology: for instance, there is no surjective continuous function $[0, 1] \rightarrow (0, 1)$.

Closed subsets The definition of open sets allows us to define another important notion in topology: that of a CLOSED SUBSET.

Definition 2.16. Let X be a space with a topology $\mathcal{T}_X$, and F be a subset of the set X . By definition, F is **closed** just in case its complement is an open set, i.e.:

$$X \setminus F \in \mathcal{T}_X.$$

Notice that this is not the same as “not being an open set.” In fact, the two concepts are independent from one another in general: many subsets are neither closed nor open, and there are always a few subsets that are both closed and open, ‘clopen’ for short. Cases in point are the empty $\emptyset$ and total X subsets of X , which are always both closed and open. Spaces in which other subsets are clopen are discussed in Section 2.9.

The definition can be reformulated in terms of neighborhoods, but it is much more natural in terms of open subsets. This constitutes another aspect which is simplified in considering closed subsets. Specifically, while an open set is any subset that is a neighborhood of all its points, a closed set is any subset F such that any point which cannot be separated from F is in fact in F (see Proposition A.6).

^uMore precisely, completely metrizable.

This definition does make intuitive sense in retrospect, in that it isolates the subsets which contain all points that they reach arbitrarily close to, or are not separated from. Put differently, a closed subset is stable by the operation of taking limits within it (whenever they already exist in the space X).

Closure A connected notion is that of **CLOSURE** of a subset. Given a subset Y of X , the closure of Y is the collection of those points that are “close enough” that all their neighborhoods have an element of Y .

Definition 2.17. Let X be a topological space and Y be a subset of X . The **closure of Y** , notated $\overline{Y}$, is the smallest closed set containing Y :

$$\begin{aligned}\overline{Y} &:= \bigcap \{F \subseteq X \mid F \text{ closed} \wedge Y \subseteq F\} \\ &= \{x \in X \mid (\forall N \in \mathcal{N}_x) N \cap Y \neq \emptyset\}\end{aligned}$$

Put differently, this is essentially the set of points that can be reached by sequences inside the subset which converge, possibly outside of the subset. An important example is found in the Sierpiński space: the point c is in the closure of $\{g\}$, but the converse is not true. I will also use this notion on other topological spaces in Section 3.3.6. This is also the only place where I use the two remaining notions. Of these, manifolds are a useful generalization of intervals and planes, but one which is not obvious. However, connectedness is a fairly intuitive notion, which can safely be skipped on the first reading.

2.9. Connectedness

In non-mathematical terms, a connected space would be described as a ‘continuous shape.’ However, mathematicians use different terms to disambiguate the notion of continuity. The term ‘continuous’ is reserved for transformations (functions), while the corresponding notion for spaces is called ‘connectedness.’ Then, **CONNECTED** spaces are defined as those spaces where the only clopen sets are the empty and total subsets. This is because if two non-empty complementary sets are open together (hence, clopen together), then they dissect the space into two components that are not connected together, because each surrounds itself, so does not sit immediately next to the other.^v

Definition 2.18. Let X be a topological space. The space X is **connected** iff whenever two open subsets partition X , at least one of them is empty:

$$(\forall O, O' \in \mathcal{T}_X) [O \cap O' = \emptyset \wedge O \cup O' = X] \implies [O = \emptyset \vee O' = \emptyset].$$

Thus, the data of open sets very straightforwardly encapsulates how connected the space is; whether it sits in one piece or not. The next question to ask is naturally what counts as a “piece” of a space, disconnected from other pieces. This is the notion of a **CONNECTED COMPONENT**, which essentially behaves as one would expect. Even spaces with non-trivial cohesion can have only points as connected components, because no non-singleton subset is connected; this is the case of the space N_ω .

^vStated differently, the existence of two non-empty complementary opens is equivalent to the existence of a clopen subset that is neither empty ($\emptyset$) nor total (X itself).

Definition 2.19. Let X be a topological space, and C be a subset of X containing some point x . The subset C is the **connected component** of x iff it is connected and no strict superset of C is connected; i.e. C is a maximal connected subset containing x .

2.10. Manifolds

Finally, a special class of topological spaces are (topological) **MANIFOLDS**, those spaces that look locally like Euclidean space in some dimension. Euclidean space in dimension n is the space where each point is described by n real numbers.^w Dimension 0 is a mere point (no real number needed); dimension 1 is the (infinite) real line (exactly one needed); dimension 2 is an infinite plane; dimension 3 is an infinite space, and so on. Locally, Euclidean space can be covered with open balls of the appropriate dimension: one-dimensional balls are open intervals; two-dimensional balls are open disks, and so on.

Then, manifolds can be thought of as deformations of Euclidean space, possibly looping back on themselves. In any case, manifolds are in a sense homogeneous spaces. This is because every point has a neighborhood which is homeomorphic to an open ball in Euclidean space, and all open balls are homeomorphic in Euclidean space of a given dimension. Thus, all points have identical-looking neighborhoods.

Definition 2.20. Let X be a topological space and n be a natural number. The space X is a **manifold of dimension n** iff for every point $x \in X$, there is a neighborhood of that point which is homeomorphic to an open ball $B_{\mathbb{R}^n}$ of dimension n :^x

$$(\forall x \in X) (\exists N \in \mathcal{N}_x) N \simeq B_{\mathbb{R}^n}.$$

Manifolds are very neatly described in terms of their connected components; here are examples of manifolds in low dimensions.

- In dimension 0, there is only one connected manifold, the singleton space. Disconnected manifolds are exactly discrete spaces.
- In dimension 1, there are essentially two kinds of connected manifolds (also called ‘curves’): the open interval and the circle (all intervals and circles respectively have the same topology). The circle is the first example of manifold which is locally but not globally like the corresponding Euclidean space. Disconnected manifolds are disjoint unions of open intervals and circles.
- In dimension 2, there are infinitely many kinds of connected manifolds (also called ‘surfaces’), but they are fairly easily described: the most familiar are the plane, the open cylinder, the sphere and the torus.
- In dimension 3, any open region of space is a manifold, e.g. a ball or a solid torus; manifolds which close up on themselves like the sphere or torus are harder to imagine, since they do not embed in 3D space. Of course, this only gets more arduous in higher dimensions.

^wFormally, this is the Cartesian product $\mathbb{R}^n = \mathbb{R} \times \cdots \times \mathbb{R}$.

^xFor the definition of $B_{\mathbb{R}^n}$, see Definition A.17.

Manifold with boundary Slightly more flexible is the notion of MANIFOLD WITH BOUNDARY: those allow for exceptional points as long as an area around them looks like the endpoint of an interval (in dimension 1), the outside (non-open) edge of a surface (in dimension 2), or the outside surface enclosing a zone of space (in dimension 3), etc. This boundary is always only one dimension below that of the general manifold. The formal definition of this notion is more involved, see Definition A.19 in Appendix A.6.

3. Temporal conceptualization of events across time

One landmark in the study of linguistic aspect is Vendler's (1957) article, which sorted predicates into different ASPECTUAL TYPES based on their combination patterns with temporal expressions. This variation between predicates was called LEXICAL ASPECT, to distinguish it from grammatical variation for presenting those predicates, grammatical aspect. This latter kind of aspect is displayed with the Past and Past Progressive sentences in (8).

- (8) a. Kim ran.
b. Kim was running.

Building on this article and the abundant literature that followed, Croft (2012) presented a way to integrate aspectual and causal descriptions of verbs, with a geometric-like representation of the unfolding of events, and a graph-like representation of their causal structure. This work, grounded in Cognitive Linguistics, aims to give a cross-linguistic account of patterns in argument realization, such as defaults and alternations. In the domain of aspect, his two-dimensional view proves quite fertile, thanks to a flexible qualitative dimension which cleans up the numerous different classifications of aspectual types identified in the literature.

3.1. Croft's system

“Lexical” aspect Crucially, Croft (2012) advocates for a view of lexical aspect which is only marginally lexical: it does not appeal to an encyclopedic lexicon listing possible aspectual types as a matter of each particular lexical item. On the contrary, his account places the emphasis on *conceptualization*, or *construal*, as one of the driving forces of lexical aspect, which is reached as a compromise between the grammatical context, the contribution of each lexical item, and other contextual factors. Aspectual types are not classes of lexical items anymore, but possible construals which exist in relative independence to the lexicon. Thus, when a token experience of the outside world is ascribed to a certain event type by an utterance, a unique construal is chosen as part of the semantic representation of the utterance, besides truth conditions. This construal contributes conceptual or discourse elements, above and beyond fact of the matter; in particular, it features an aspectual component, its aspectual type. I refer to Croft (2012, §1.4) for a more in-depth discussion of the concept.

(Aspectual) frames More specifically, Croft's construals involve the cognitive-linguistic notion of frames introduced by Fillmore (1982). A FRAME is a structure surrounding the concept brought up by a word, which is essential in understanding the concept in question. It can consist in coordinate, contrasting concepts, like the concept of SEA which belongs to the same frame as LAND; or simply in additional elements than those strictly meant by the concept, like the frame of BUY, which evokes a transaction with the presence of MONEY, but which is backgrounded relative to the other participants. This backgrounding phenomenon is potentially related to the formal-semantic notion of presupposition, meaning those entailments of a sentence that are preserved under operators like negation and thus are required to hold for the sentence to be appropriate. The term ‘presupposed’ is used differently in frame semantics than in formal semantics however, and it usually taken to apply to any concept evoked by another. That being said, it seems in fact quite plausible that such backgrounded elements directly lead to presuppositions, when the words are used inside sentences.

The frame-semantic point of view seems quite appropriate for aspect: understanding the aspectual verb *start* requires understanding the event whose inception it refers to, and whose existence is certainly entailed in sentences. Croft proposes that the aspectual representation of all utterances involves

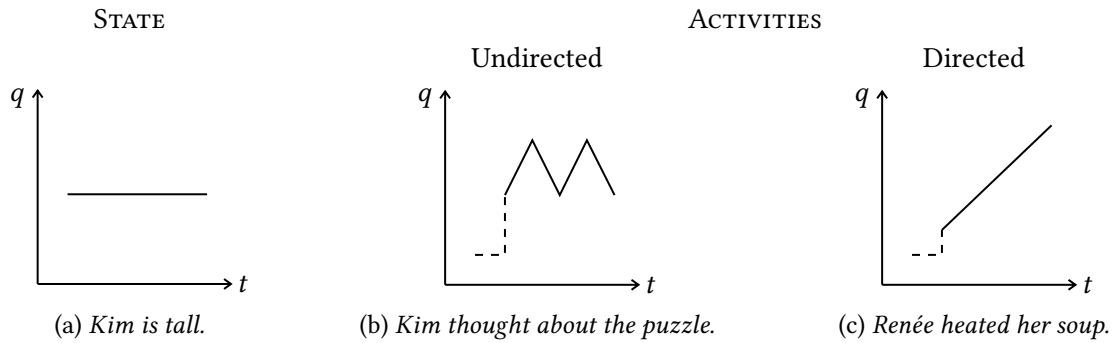


Figure 12: A state and two events (which are both activities)

a temporal–qualitative frame whose subparts can be focused, much like *start* might PROFILE (highlight) an inception stage.

How events unfold This temporal–qualitative frame serves to describe how an event is construed as unfolding over time, “the sequence of qualitative states that characterizes [the associated] event type.” I call this the **TRAJECTORY** of the aspectual construal, which takes place over time, the t dimension, and within a collection of qualitative states that Croft calls the q dimension. The trajectory is simply a particular way of navigating that dimension across time, and makes up the meat of Croft’s two-dimensional aspectual representations. The t and q dimension are part of the frame but cannot be profiled; it is the trajectory which features a profiled part and a backgrounded, presupposed part.

Aspectual classification In Croft’s representations, trajectories are schematized, using only a few basic graphical patterns which reflect features of the q dimension. These patterns in fact partially motivate his aspectual classification, though many of his distinctions were already established in the literature. I summarize the classification here with example sentences, but without recalling the underlying linguistic arguments. Most basic is the Vendlerian distinction between **STATES** like (9), which feature a flat trajectory reflecting a constant mode of being, and **EVENTS** whose profiled part involves changes along q .

(9) Kim is tall.

Within events, zigzag lines represent undirected activities such as thinking (10a), understood as cyclic changes between two or more modes of being, while tilted slopes represent directed activities with a continuous change in one qualitative direction like gaining heat (10b): Figures 12b–12c.

- (10) a. Kim thought about the puzzle.
b. Renée heated her soup.

For both of those last two types, a result state can be added to form telic predicates called **ACCOMPLISHMENTS**: **NONINCREMENTAL** in (11a) and **INCREMENTAL** in (11b).

- (11) a. Kim solved the puzzle.
b. Renée heated her soup up 5 °C.

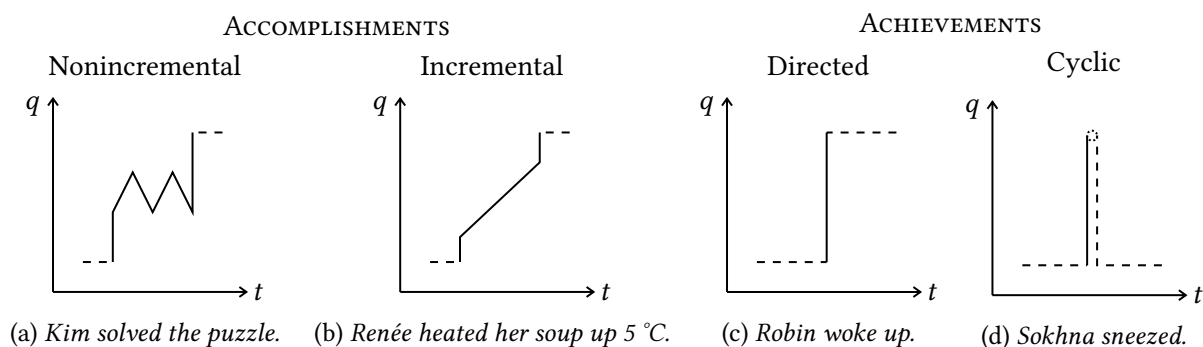


Figure 13: Other events

The corresponding representation features a final rest stage in Figures 13a–13b respectively. Note that, by convention, types which involve directed one-dimensional change are drawn with increasing trajectories, and result states are drawn above other points in the q dimension.

Next to these three kinds of extended trajectories (flat, zigzag or slope), Croft also appeals to instantaneous transitions to represent ACHIEVEMENTS, which are punctual changes. If a new state results like for (12a), they are called DIRECTED, while if they revert back without any salient change in the state of the entity, like (12b), they are CYCLIC (also called SEMELFACTIVE in the literature): Figures 13c–13d.

- (12) a. Robin woke up.
b. Sokhna sneezed.

Finally, Croft mentions the possibility of differentiating states based on their extension on the t dimension: they can be (construed as) POINT-LIKE like (13a), TRANSITORY, like (13b), IRREVERSIBLE like (13c), or PERMANENT like (13d).

- (13) a. It is 5 o'clock.
b. Kim is awake.
c. The window is shattered.
d. Kim is tall.

Irreversible ('acquired permanent') and permanent states have an arrow in their representation to reflect this permanence, as in Figure 14. Since these distinctions can be made for states, they can also apply to achievements, which are simply the analogues of non-permanent states which profile the inception stage instead of the result stage. Under this conception, to point states correspond cyclic achievements, and to transitory states correspond reversible achievements; both types of achievements mentioned above. Lastly, irreversible achievements similar to (14) correspond to irreversible states, but I do not represent them here.

- (14) The window shattered.

Appeals of the system The view of lexical aspect that emerges from Croft's aspectual types is a quite organized one. Only three types of trajectories are present, which are put to full use in a coherent, symmetrical system. This brings out especially clearly the analogies between activities and

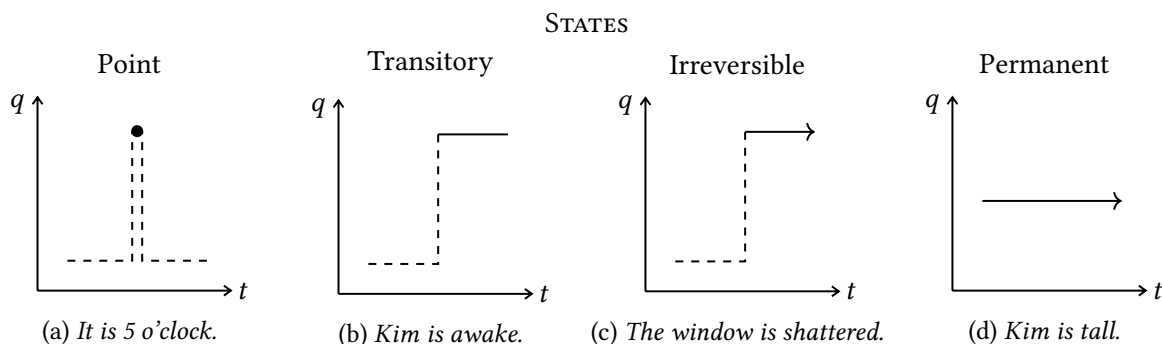


Figure 14: Different kinds of states.

accomplishments, as well as between states and achievements. Though the affinities between these different types are well-known, the cross-cutting distinctions with directed and undirected process for the former and punctuality or reversibility for the latter are new, as far as I am aware.

The system also makes it especially easy to change construal within common aspectual pairs. States are obtained from achievements by a simple change in the profiled part of the representation, which is one of the available parameters among aspectual types of the theory. Achievements are obtained from accomplishments by adopting a coarse-grained, compressed view on both dimensions. Both types of change were already showcased in the introduction. Because the transitions are so easy to express, this constitutes a first step towards a **compositional theory of lexical aspect**. Furthermore, both processes raise the possibility that some kind of **attention** could play a major role in the conceptualization of events, just like is well-established in lower cognition, such as the visual domain. The possibility that this computationally reasonable principle could be an overarching one in cognition is quite tantalizing.

Finally, this representation fits inside a wider system which also represents the causal structure of events, as one of the factors in argument structure. The aspectual part of the system is well integrated: the causal graph rests on specific stages of the aspectual trajectory, and a procedure is described to derive the global construal of a complex event from the local construals of its subevents.

All these aspects of Croft's proposal make it a very promising system, with real potential to illuminate the study of event conceptualization if approached with the goal of promoting it to a real theory. A few issues emerge about the system, however, surrounding the qualitative dimension.

3.2. Drawbacks and possible solutions

Meaning of q points First, though three kinds of trajectories are present to suggest different underlying spaces on the q dimension, the states that this dimension features are sometimes left underspecified. Those states are not ignored for aspectual-type purposes, though: Croft states that “it is the q scale, no matter how it is derived ..., that determines the aspectual potential of the predicate” (p. 74).^y This suggests that some notion of equivalence of the qualitative scale is crucial in defining aspectual types; but no such notion is defined. On the other hand, in differentiating the accomplishment-like result verbs as in (15a) and the activity-like manner verbs as in (15b), he writes that “the difference between [those] verbs is the presence of a directed change in the verbal profile, not a difference in the

^yHis example involves *mowing the lawn*, seeing the incremental theme of this accomplishment in one of two ways. The usual way is as a *mereological theme* where bigger and bigger parts of the lawn are fully mown. The other possibility, which requires different technology, would be as a *property theme* where the height of the entire lawn is progressively reduced. In both cases, the result state is the same.

states on the q dimension” (p. 301). But the alleged identity of qualitative states in the two cases is in seeming contradiction to their different aspectual behavior, which suggest non-equivalent scales.

- (15) a. The hummingbird flew into the nest.
b. The hummingbird flew around.

I believe that we can clarify this by differentiating two characteristics of q states that are at play here: their event-type meaning and their aspectual-type meaning. Croft presumably means that it is the event-type-specific interpretation of the states that is identical for manner and result verbs: *flying into the nest* and *flying around* are both instances of the FLYING manner of motion. On the other hand, aspectual-type meaning reflects the involvement of a q state **with respect to the unfolding of the event**, irrespective of how each state is interpreted in the situation. From this perspective, aspectual meaning involves progression towards a terminus in the former example, and no such directedness in the latter: as different aspectual types, I contend that a clear difference should be represented in the q dimension. In fact, I would like to raise the possibility that **event-type meaning of q states should be entirely irrelevant to aspect**, and that a definitional factor for aspectual types, next to the trajectories themselves, is the global “aspectual cohesion” of the collection of points. To stress the importance of this collection as a whole, I introduce the concept of **conceptual space**, with the idea that the topology of that space should reflect how the points can be involved as the event unfolds. Aspectual equivalence of spaces then simply amounts to topological equivalence, two spaces being equivalent if the cohesion among their points is the same. Under this view, qualitative states and their aspectual-type meaning are integral to the construal of an event’s unfolding, so that **trajectory and conceptual space go hand in hand to define aspectual type**. I return to this in the next section.

Besides the nature of the states, certain representations feature specific ambiguities about the composition of the q scale, or conceptual space.

Discontinuity One ambiguity is about the continuity between the points. For achievements, Croft proposes that discontinuities are present in q , and this does seem tenable in the case of cognition (*notice*) and possession (*obtain*) predicates. However, if location is recorded on the q dimension without any adjustment, then we cannot contend that a locative verb like *reach* involves discontinuous change as such. Yet, Croft uses a discontinuous trajectory for all achievements. In Section 3.3.2, I argue that the conceptual space is really **isomorphic in both subclasses of verbs**, namely the Sierpiński space showcased in the introduction. This space allows intuitively discontinuous trajectories without actual, formal discontinuity.

Activities Second, Croft holds that undirected activities should involve at least two states that the entity goes back and forth between, or at the very least that cannot be ordered (p. 61).^z Nevertheless, he draws a zigzag for those activities that seems to consist of extended directed segments which turn around at each iteration, suggesting that a linear continuum does exist when this does not seem implied by his formulation. Though at a broad-stroke level this makes intuitive sense, it is not clear at all whether these transitions are supposed to be instantaneous, or if the transitions should look different should there be more than two of those alternating points. Prioritizing again aspectual-type meaning of the q scale, I argue that a different approach should be taken to undirected activities, which eschews the event-type meaning of having alternating points in the conceptual space and instead focuses on having a **homogeneous space** with no manifest progression, whether one-dimensional or toward a result state. This is discussed in Sections 3.3.5 and 3.4.1.

^zOf course, a one-dimensional representation will naturally suggest that the points can be ordered, but this is taken to be an artifact.

Trajectories Croft presents the different kinds of phases that can make up each trajectory, listing constant segments as well as instantaneous, directed, and undirected change. But, due to the one-dimensional axis representation, it is not obvious what is to be made of other conceivable trajectories. One could imagine circles within a two-dimensional conceptual space, or simpler, two contiguous phases: one with directed change and another being an extended state. Such phases seem to always be separated by an instantaneous transition, but no principled reason why this should be the case is given. Though I do not provide a solution to this problem, in Section 3.4.2 I specify certain conditions on trajectories that bring us closer to regimenting this aspect of the system.

Change of granularity Finally, because the q dimension is left vague, some operations which are appealed to in explaining aspectual potential of predicates cannot be described precisely. Though Croft states that scalar adjustment occurs simultaneously in both t and q dimensions, this cannot be the whole story. While taking a coarse-grained view on a continuous accomplishment scale understandably yields a two-state achievement scale (see Figures 2–3), the same operation on the two-state scale of cyclic achievements is proposed to change it into a more precise activity scale (change from fine-grained *He coughed* to coarse-grained *He is coughing*).^{aa} Furthermore, it is not self-evident why taking a fine-grained view on the activity construal of a state-like predicate in the Progressive (16a) results in a constant state (16b), as Croft suggests by representing a permanent state construal.

- (16) a. She is resembling her mother more and more.
 b. She resembles her mother.

Though he is right that a fine-grained view will remove a curve's curvature, a zoom operation that happens in both dimensions and thereby preserves relative scale will not remove slope: a mountain side seen up close still features a gradient. For this example, it seems more appropriate instead to consider that the adjustment occurs in the t scale only, or to analyze the operation in an altogether different way. After presenting an approach for changes of granularity in Section 3.6 I opt for the latter, considering that **the permanent-state construal is really a coarse-grained view of the directed activity**. I also propose a **more precise, formal definition** of scalar adjustment to account for changes in aspectual type.

3.2.1. Formalization

Interpretive layer In the preceding section, I have sketched some changes in interpretation, which should be summarized here, introduced by my formal account compared to Croft's. I aim to draw an aspectual typology of eventualities^{ab} by defining each aspectual type as **a sequence of qualitative states within a specific conceptual space**, which I call a PHASE SPACE. I also call the points of the phase space PHASES, using the term STAGE instead to refer to each leg of a trajectory. Such a sequence is dependent on the phase space, which must therefore appear in the definition of the aspectual type. The points and topology of this conceptual space are part of the **aspectual frame** of the aspectual type, so that distinctions and topological separation between these points are only assumed inasmuch as they matter for aspectual description. This means that there is no universal conceptual space within which each individual could be placed at any instant, and conceptual spaces are not specific to each predicate either. Instead, each aspectual type has a unique space associated to it, which is brought to mind for any sentence with that aspectual profile. Though (17a) might have a conceptual frame evoking all other citizenships, its *aspectual frame* simply features one point.

^{aa}The three states in question are the rest state, as well as the other two states that the entity alternates between.

^{ab}In the sense of Bach (1986), encompassing both states and non-states (which are "events" in the narrow sense).

- (17) a. Nadia is Polish.
 b. Nadia became Polish.

By contrast, (17b) features a two-point space, one point being the outcome of the change and the other point a preceding nondescript “rest” state.^{ac} Since the difference is aspectually relevant, a distinction is made. However, in the accomplishment (18), it is only Nima’s progress in mowing the lawn that is recorded in an interval-like space, no matter whether the progress is along the surface of the lawn (mereological) or the height of the lawn (property).

- (18) Nima mowed the lawn.

Similarly, any accomplishment will be represented with the same scale, position on that scale representing progress along the event.

The sequence of qualitative states is taken to apply to a salient individual in the situation described by the sentence, just as Croft (2012) suggests, before he extends the description to causal structure: then, he uses different states for each participant. Each step in the sequence is like a timeslice of the individual as it traverses the phase space. The sequence is formalized by a continuous function, called a **trajectory**, going from a time window (the interval space $I = [0, 1]$) into the conceptual space. More specifically, a **canonical** trajectory is chosen to represent the unfolding of the event: for instance, the canonical trajectory for an accomplishment or achievement should be a culminating one, which reaches the culminating point of the phase space and stays there for some time. This contrasts with the reverse trajectory which starts at the culminating point and ends up at the rest point, or any constant trajectory, which would not qualify as “culminating.”

The points of the time window I will stand for a continuum of *relative* time instants, with no precise meaning assigned to the distance between the real numbers that represent the points. This fact allows this window to represent any timescale of relevance (for instance, logarithmic). Because of this indifference to actual time values, I will not consider simple trajectories but rather **equivalence classes of trajectories**, so that speed is not relevant. Similarly, only the relative point of progress in the conceptual space is taken to matter, so that topologically equivalent trajectories are considered to stand for aspectually equivalent conceptualizations. To do this, I consider trajectories *up to reparametrization* (any increasing, bijective relabeling of the time points) to neglect speed, and *up to self-equivalence* (any automorphism of the space, see Definition 3.2) to neglect absolute point of progress; see Section 3.4.1.

3.3. Spaces and sample trajectories

I now present the topological objects that I will use as phase spaces, saying a few words about their respective properties and what trajectories in them look like. All of them feature one open point, which can be understood as a nondescript REST PHASE and will be taken as the initial point of all canonical trajectories. I do not introduce any notion of profiling until Section 3.4.2, so for now all trajectories are drawn with solid lines. Full circles represent the actual value of the function at the t -axis point below, while open circles indicate a limiting value around that point.

^{ac}The nondescriptness might not stop subsequent inferences from being drawn about the rest states, especially for achievements: here, the preceding stage could be deduced to be one of non-Polishness, if one is aware that Poland tolerates dual citizenship. For ‘became Japanese,’ one can conclude that the rest state is a disjunction of all other citizenships and statelessness, since Japan does not allow dual citizenship. However, nothing in the aspectual representation suggests this; the reasoning I just described would be entirely independent from the aspectual representation in itself.

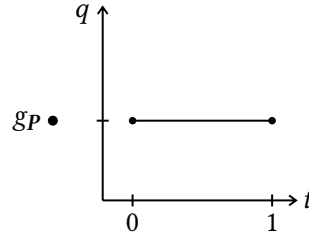


Figure 15: The singleton space P and the only trajectory within it, used for **permanent states**

3.3.1. Singleton

First, the **singleton space**, written P for ‘point,’ has just one element g_P , which coheres to itself in the only possible way: its only neighborhood (and the only non-empty open subset) is $\{g_P\}$, i.e. the whole space P itself. Because there is only one point, and because it is also a trivial space (see p. 17), any function into P , whatever the domain, will be constant and continuous. In particular, there is only one trajectory into it, that is only one (continuous) function which has I as domain and P as codomain. That function is the one which maps any time instant t to the only point g_P , and is represented in Figure 15.^{ad} Any function can be described like above, by specifying its action on each input point: this is written as ‘ $t \mapsto g_P$ ’ and read ‘ t maps to g_P ,’ where the right side may depend on the free variable on the left side.^{ae} The singleton space can profitably be used to describe **permanent states**, as there is no other point that the entity could possibly move to, or origin for the point to move from.

3.3.2. Sierpiński space

Second, the **Sierpiński space** S has two points: one open point which I will label g and one closed point c , which can be conceived of as having an additional property. Because of their respective openness and closedness, they cohere together in a particular way, so that any function into S corresponds to picking out an open subset of the domain, all of whose points are sent to g ; the remaining points which form a closed subset are all sent to c . Any open subset (or its closed complement) thus corresponds to a bona-fide continuous function into S . In the special case of I , a closed subset corresponds to a finite or empty union of elementary closed subsets which are either isolated points, accumulation points (such as ω in $\mathbb{N}_\omega$), or closed intervals. Example trajectories corresponding to each kind of elementary closed subset are given in Figure 16. The trajectory with an accumulation point does not seem to be useful or even sensible (see Definition A.4 for an explicit formula and Section 3.4.2 for further comment), but the remaining trajectories with isolated points and closed intervals, especially the first and last ones, are essentially point and transitory states respectively. Thus, they provide justification for considering S to represent **transitory states**, both extended and point-like. The corresponding **achievements** (directed or cyclic) are also available under different profiling of the same trajectories; I will get to profiling in Section 3.4.2.

Discontinuities and disconnectedness Notice that the space S is both connected, in the sense of Section 2.9. What happens here is that the abstract notion of closeness of general topology allows points that are distinct but still immediately adjacent and accessible.^{af} This does lead to intuitive

^{ad}The graph for the function has dots on either side to make explicit the fact that the function has a value at 0 and 1, instead of being merely defined on the open interval $(0, 1)$.

^{ae}This is a notational variant of λ -abstraction ‘ $\lambda t. g_P$,’ slightly more flexible in that it allows to specify a value only for specific members, unlike λ -terms which describe an entire function.

^{af}This could not have happened in a general metric space: the only two-point metric space has a discrete topology (see p. 17).

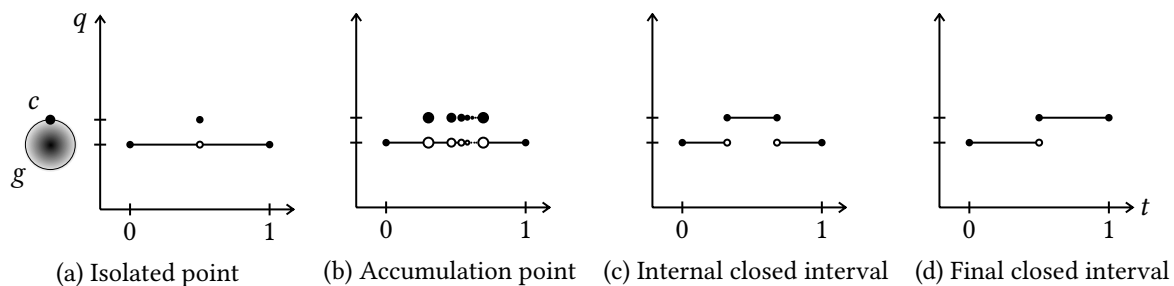


Figure 16: The Sierpiński space S and four trajectories within it, each corresponding to a closed subset of I which is sent to c . These could be used for **punctual states/cyclic achievements**, and **transitory states/reversible achievements**.

discontinuities when imperfectly represented on a piece of paper (where distinct points are never adjacent), since the trajectory can move in an instant from one point to the other. However, recall that topologically this is not a discontinuity in any meaningful sense, as the trajectory complies with all conditions required of continuous functions: only transitions respecting each point's neighborhoods are allowed. This is in stark contrast to actual **disconnected** spaces, such as discrete ones, in which no continuous trajectory exists between distinct points whatsoever; joining distinct points requires breaking continuity. This is why I use such a space for permanent states. Note that I do not assign any specific aspectual meaning to transitions from an open point to a closed one as opposed to the reverse situation; I just regard it as an artifact required by the formalism, inasmuch as open points are chosen to represent rest states.

This space allows us to bring all achievements under one umbrella, regardless of distinctions in semantic type, such as those of Dowty (1979, p. 68). Admittedly, at an intuitive level many predicates of cognition (*realize*), possession (*lose*), and change of state (*explode*) could receive a “discontinuous” space on account of their harder construal as accomplishments in the progressive. Though other changes of state are more amenable to such construals and readily allow intermediate states (*die*, *fall asleep*, *reach*), this seems far from categorical, and more dependent on conceptual preferences for certain predicates than hard potential. In light of my focus on aspectual-type meaning of q states, I believe that **this variation between event types should not necessarily be recorded in the q dimension**. Instead, all achievements make use of a space in which one state is instantly accessible from the other, instead of different spaces with similar trajectories in them. Then, all achievements could undergo the conceptual-space manipulations to be discussed in Section 3.6, though with more or less ease depending on the actual predicate; since some conceptual correspondence should occur between q states and event-type events.

This does introduce a topological asymmetry between the open rest state and closed result state, which could be problematic due to “symmetric” achievement verbs such as *melt/freeze* or *open/close*. But given that only aspectual meaning is given to points in conceptual spaces, no such problem arises. Even so, if our current interpretation was enriched with event-type considerations and conceptual differences between open and closed points, a possibility could be that such pairs do not in fact have perfectly symmetrical conceptual counterparts.

Irreversibility and transience A notable type posited by Croft (2012) that is missing from my account is that of irreversible achievements, which can be set apart from reversible or resettable ones with iterative adverbials (Talmy 1985):

- (19) a. He fell three times.
b. *He died three times.

A comparable argument to distinguish transitory and acquired permanent states could be the ability to occur with *still* (Klein 1994, p. 85).

- (20) a. The door was still open when I came back.
b. *The quagga was still extinct when Burton first came to Africa.

However, it is noteworthy that no analogous “irreversible accomplishment” class seems to have been proposed (with members like *eat the sandwich* or *raze the bridge*). Though alternative construals allow irreversibility to be fluid for each predicate, this is subject to considerable variation depending on the imagined, possibly imaginary, context.^{ag} The fact that this distinction rarely matters in the system, and involves a merely diacritic representation, should raise doubts as to its relevance. Indeed, it is not clear that this difference bears on the unfolding of events per se, and not instead on general reasoning mechanisms that make sentences such as (21) unacceptable on logical grounds.

- (21) ?Vova did a handstand on his feet

Therefore, I suggest that these facts **might not pertain to aspectual construal**, and the topological analysis I propose for this domain does not enforce any notion of irreversibility. Indeed, any continuous trajectory within a conceptual space can be entirely reversed because of the topological symmetry of the time window I .^{ah} Thus, while permanence (i.e. mutual inaccessibility between states) can be encoded naturally with a discrete space, irreversibility or one-way change cannot be forced with topology in any phase space.^{ai} I argue that what predicates are potentially irreversible is a matter of world knowledge. Occasional construals of those predicates as reversible could be partly explained as non-aspectual construal (e.g. construing participants as types instead of tokens), or as the matter of another reasoning module entirely.

The predictions from this stance in terms of grammar are hard to cash out because aspect is really a covert category. It may be something like the following: **there are no lexical items which are exclusively irreversible**, and cannot occur with otherwise productive iterative constructions (whether lexical or derivational). In particular, there should be **no derivational marking of irreversibility** (either for achievements or for states). The only marking that is allowed, like the adverbial *forever*, should be merged above Tense.

3.3.3. Closed interval with rest state

The closed interval with a rest state, which I represent with boldfaced I , is exactly like a gluing-together of the Sierpiński space S with the closed real interval $[0, 1]$, along the two points c and 0 (which become the same point in I). I notate its points with a subscript, to distinguish them from their analogues in other spaces: g_I is the open rest point, 0_I is its adjacent closed counterpart standing in for the initial state of the event, and 1_I is the opposite, result point. Trajectories within this space are just like continuous functions into a real interval, except that there may be jumps between g_I and 0_I just like

^{ag}Dowty (1979, pp. 184–185) admitted to the same variability regarding his cross-cutting agentivity distinction among predicates.

^{ah}This is because the function $I \rightarrow I, t \mapsto 1 - t$ is a function, and precomposing by it exactly yields a “reversed” trajectory. Indeed, composing two continuous functions always gives a continuous result (as can easily be checked with the definition in terms of opens; see Definition A.7).

^{ai}Unless a very different system is imagined where the time window itself is asymmetric.

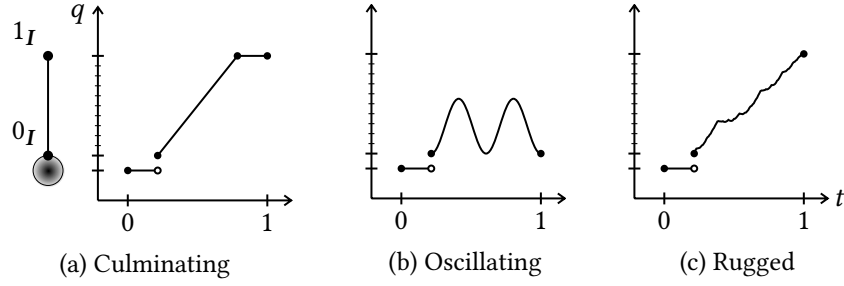


Figure 17: The closed interval with a rest state I and three trajectories within it, used for (incremental) **accomplishments**.

in S . Since no notion of directedness is enforced by the topology, a trajectory might also go back and forth between two points (traversing the intervening continuum) like Figure 17b, or be rugged with small, occasional dips down like Figure 17c – as when running a tape that is occasionally rewound by a few seconds. Therefore, these dips must happen continuously, true jumps being disallowed.

To represent **accomplishments** using the space I , I choose to isolate those trajectories, which I call ‘culminating,’ that increase along the interval, until they stay for some extended time at the result state. An example is given in Figure 17a. Both conditions are important: some increasing functions only reach 1_I at the very last moment of the time window, namely 1. Culminating trajectories correspond quite closely to Croft’s representation of directed accomplishments given in Figure 13b. However, they do not feature a final jump from the body of the accomplishment to its result state, unlike the initial jump which is much like that of achievements. Thus, assuming that the initial jump is highlighted by *start*, while the final transition into the culminating state is highlighted by *finish*, could be seen as introducing asymmetry between those two aspectual verbs’s representations. I argue that this apparent asymmetry does not necessarily matter: there is still a qualitative formal difference between the part of the trajectory that lies strictly below 1_I , which can always still proceed ahead, and that which has reached 1_I , which finds no “more complete” state to move into.^{aj} In fact, as we shall see in Section 3.6 a coarse-grained view of accomplishments can represent this qualitative change exactly as an achievement. Then, using this coarse-grained view for *finish*, and an analogous one for *start*, the asymmetry becomes quite minimal, as both verbs would require manipulation of the space to be expressed.

3.3.4. Half-open interval with rest state

This space is similar to the closed interval I , but does not feature any culminating state, so that it instead results from gluing together S and the half-open $[0, 1)$. I represent it as D , and use it for **directed activities**. Trajectories within it are just like those within I , except there is no hope for them to culminate given that the space is not compact: there are increasing sequences in it that do not converge to any point. I represent the fact that this space has no “final element” by putting ticks on the q axis that appear closer and closer as they go up, but which never converge inside the space. Appropriate trajectories for directed activities could be those that jump from g_D to 0_D and increase until the end of the time window without stopping anywhere: one such trajectory is found in Figure 18. If we consider the space *up to automorphism* (see Definition 3.2 on p. 43), the endpoint of the trajectory does not actually matter so long as it is neither 0_D or g_D : any two of the other points will have identical neighborhoods, and each can take the other’s place under appropriate (automorphic) stretching or squishing of the interval. Therefore, when trajectories are considered up to reparametrization and

^{aj}However, because I do not enforce any irreversibility, the tape can indeed be ‘rewound’ even once the result state has been reached.

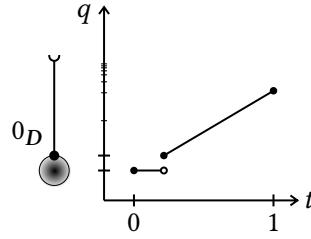


Figure 18: The half-open interval with rest state D and an increasing trajectory within it, used for **directed activities**

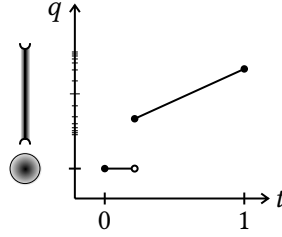


Figure 19: The open interval with rest state U and a simple trajectory within it, used for **undirected activities**

self-equivalence, all increasing trajectories become equivalent, no matter what their endpoint is.

3.3.5. Open interval with rest state

This space U is somewhat different from all preceding ones in the following sense: all other points can be reached instantaneously from the rest state, not just the “initial” state. Thus, it acts like an open interval $(0, 1)$ where a generic point g_U has been added. Formally, this means that all points have g_U as a member of all their neighborhoods (equivalently, g is an element of all non-empty open subsets). Graphically, I represent this by drawing the open point slightly apart from the other ones, and showing a distinctive, imaginary halo around the open interval to evoke the idea that all points in that interval are within reach of the open point. Notice that in this case, the open interval “ $(0_U, 1_U)$ ” is not actually an open subset of U ; however, all subsets of the form $(a, b) \cup \{g_U\}$ are indeed open.

This space is used for **undirected activities**, and all the kinds of trajectories outlined for I are also available here, keeping in mind that there are no edges to the interval. However, one especially simple kind of trajectory would be one that moves along the interval in one direction and stops at one point; this is represented in Figure 19. This choice might be surprising as it involves change which is in some sense directed. However, there is no privileged direction in this space since there are automorphisms that reverse the open interval entirely. Therefore, unlike the preceding interval-like spaces, it does not really make sense to talk of “directed” change, which is precisely why I choose this space for undirected activities.

3.3.6. Conditions on phase spaces and illegal spaces

All phase spaces listed above satisfy several conditions, which I formulate here. I will not comment on these conditions further, but they represent one first step towards a precise account of what a phase space can be in full generality. Further conditions should doubtless be added, so that the list of spaces I present becomes an exhaustive one.

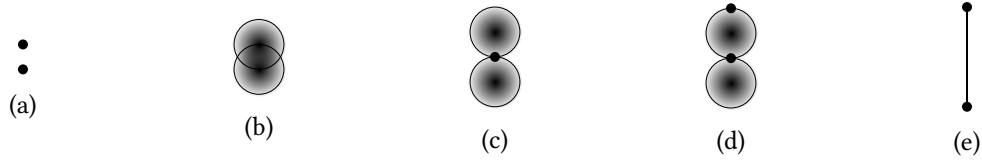


Figure 20: Spaces which do not have exactly one open point

Open point First, there should only be one open point in the space, which I will always call g_X or g for short.

(22) **Uniqueness of open point:**

$$(\exists! g \in X) \{g\} \in \mathcal{N}_g$$

Second, the space should be connected. This excludes non-singleton discrete spaces such as Figure 20a, but also trivial spaces in which no point is open, like Figure 20b (already present in Figure 4c in the introduction). Other examples include the slightly more complex spaces in Figures 20c and 20d, as well as mundane intervals (Figure 20e).

Active subset Second, the complement $X \setminus \{g\}$ of the open point g , which I call the **ACTIVE SUBSET OF THE SPACE**, should be a **connected manifold, possibly with boundary**.

(23) Conditions on the active subset $X \setminus \{g\}$

a. **Manifold with boundary:** see p. 28 and Definition A.19

b. **Connectedness:**

$$(\forall O, O' \in \mathcal{T}_{X \setminus \{g\}}) [O \cap O' = \emptyset \wedge O \cup O' = X \setminus \{g\}] \implies [O = \emptyset \wedge O' = \emptyset]$$

This prevents the active subset from being disconnected, as in Figure 21a. The manifoldhood condition also leaves out many more exotic spaces: the “longer Sierpiński spaces” in Figures 21b and 21c, but also spaces with branchings like Figure 21d. Within manifolds, different cases emerge, described on p. 27 and which I recapitulate here. First there is the possibility of the active subset being empty, and satisfying conditions (23a) and (23b) trivially: this gives the singleton space P . The only non-empty, zero-dimensional, connected manifold is a point, and this case exactly corresponds to the space S . As for connected one-dimensional manifolds, there are two without boundary: the open interval and the circle; and two with non-empty boundary: the half-open interval and the closed interval. The first two cases correspond to the space U , and a circle with rest state which I do not use. Additional conditions might be placed on the active subset, like that of being contractible: this would prevent any “holes” in the topological space, and in particular a circle with rest state. The latter two cases yield the remaining two spaces: D and I .

There remains the possibility of higher-dimensional manifolds, such as planes, spheres, hyperspheres, etc. I propose that phase spaces constructed from these are **simply not part of humans’ aspectual capacity**.

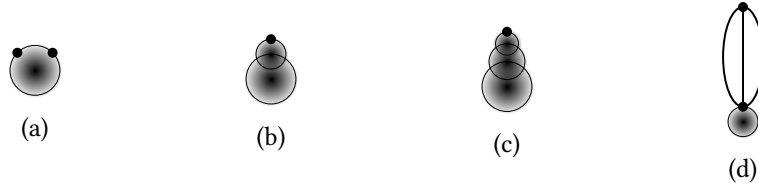


Figure 21: Spaces for which the complement of the open point is not a connected manifold

Initial subset Finally, the topology on the active subset is not enough to determine the topology on the entire space uniquely: for this, we need conditions on what the points that sit in the closure of the open point g should be. I call the collection of these points to the exclusion of g , formally $\overline{\{g\}} \setminus \{g\}$, the **INITIAL SUBSET** of the phase space. By topological properties and the previous conditions, it is closed, and non-empty when the active subset $X \setminus \{g\}$ is non-empty.^{ak} I further require that **the initial subset be a connected manifold without boundary**. For the spaces seen so far, this subset should then be a zero- or one-dimensional manifold: in the first case, this is a single point; in the second case, this is either a circle or an open interval. However, topological properties then give no choice as to the initial subset. When the active subset is a manifold with non-empty boundary, the initial subset must be a single point.^{al} if we also require this point to be in the boundary of the active subset, this yields exactly the two spaces I and D . When the active subset is a manifold without boundary, the initial subset must be either a point or the entire active subset.^{am} For the Sierpiński space S this makes no difference, but this leaves two choices for the open interval, of which I choose to only keep U .

Extra, unregimented conditions Note that for the spaces with intervals, I have not excluded with an intensional condition the cases where the initial subset is an interior point of the active subset; instead, I make it a boundary point for I and D , and make it the entire active subset for U . I do not have a principled justification for either of these choices. Further explanation, or amendment of the phase spaces to reach a tamer set of constraints, will have to await future research.

3.4. Trajectories: equivalence classes and profiling

3.4.1. Reparametrization and self-equivalence

Recall that a trajectory within X is any continuous function $p : I \rightarrow X$ from the closed interval I into the topological space X . However, I mentioned in Section 3.2.1 that the time window is meant to be flexible, in the sense that the distance between the time instants does not matter, only their relative order does. Thus, the precise moments at which a phase trajectory takes on a specific value will be irrelevant for conceptualization, so much as the sequence of phases the trajectory prescribes. More radical yet, I propose that the precise points visited by the trajectory should not matter for aspectual conceptualization; instead, the general movement of the trajectory within the space should. Formally, this is done by considering trajectories UP TO REPARAMETRIZATION OF THE INTERVAL and UP TO SELF-EQUIVALENCE OF THE PHASE SPACE.

^{ak}Indeed $\overline{\{g\}}$ is closed by definition, and $\{g\}$ is open so its complement $X \setminus \{g\}$ is closed, too. Then, $\overline{\{g\}} \setminus \{g\}$ is closed as the intersection of two closed subsets, since $\overline{\{g\}} \setminus \{g\} = \overline{\{g\}} \cap (X \setminus \{g\})$.

^{al}To see this, notice that the initial subset is both closed and a subset of the active subset, which is connected. The initial subset cannot be a circle given the topology of the active subset, but it could be an open interval. In that case, the initial subset would be a clopen subset of the active subset, so either total or empty. Neither option is possible: in the first case the initial subset is not a manifold without boundary, and in the second case then the phase space is not connected. This only leaves the case where the initial subset is a single point.

^{am}In higher dimensions, other configurations are possible.

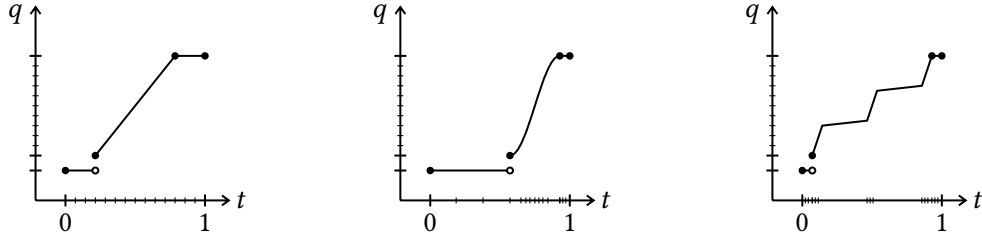


Figure 22: Three reparametrization-equivalent trajectories in I , with ticks in the t axis representing the reparametrizations

Definition 3.1. A **reparametrization** of the interval I is an increasing bijection $\phi : I \rightarrow I$. Any such function is necessarily continuous (and so is its inverse). Two trajectories $p, p' : I \rightarrow X$ are **equivalent up to reparametrization**, written $p \equiv_I p'$, if there is a reparametrization $\phi : I \rightarrow I$ such that $p' = p \circ \phi$.^{an}

Definition 3.2. An **automorphism** of the phase space X is a continuous function $f : X \rightarrow X$ which can be inverted by a continuous function: there exists another continuous function $g : X \rightarrow X$ which is an inverse to f , i.e. such that $g \circ f = \text{id}_X$ and $f \circ g = \text{id}_X$. In a sense, this is a relabeling of the points of X which respects the topology; it is simply another way to see X . (In fact, a reparametrization is a kind of automorphism of I .) Two trajectories $p, p' : I \rightarrow X$ are **equivalent up to automorphism**, written $p \equiv_X p'$, if there is an automorphism $f : X \rightarrow X$ such that $p' = f \circ p$.

See in Figure 22 three reparametrization-equivalent trajectories, the first being the usual one. The second one has initial stretching, middle deformation and final compressing of the t axis, while the third one has alternating compressing and dilating of the axis.

As for automorphism-equivalent trajectories, a real difference with reparametrization-equivalence appears in trajectories that do not visit the whole space, as with the sample trajectories I presented for the spaces D and U . Figure 23 presents three equivalent trajectories: the second one is obtained via heavy compressing of the open interval, while the other one follows from dilating and reversing the orientation of the interval. Other, equivalent trajectories could be obtained by “sliding” the middle of the interval (as addition does on the real line). This notion of equivalence is why the one-dimensional space U can be used for undirected trajectories: although the activity does progress along a continuum, there is no sense in which it follows any directed change. Therefore, **undirected activities are profitably represented using a homogeneous space**.

In the end, considering trajectories up to equivalence implements indifference to scale: instead of recording exactly where the trajectory goes at each instant among an uncountable infinity, it is enough to know what stages the trajectory goes through, however brief. This puts more focus on the topological properties of the time window and the phase space, since the labels used do not matter, only the general movement of the trajectory does. This allows undirected activities to be represented with the “homogeneous” space U .

3.4.2. Profiling

In order to profile certain stages of a trajectory (or its equivalence class), we should define what counts as a stage of that trajectory in the first place. This idea is already present in Croft (2012): each stretch of

^{an}Recall that if $f : B \rightarrow C$ and $g : A \rightarrow B$ are two functions, their successive composition, mapping $a \in A$ to $f(g(a)) \in C$, is written as $f \circ g : A \rightarrow C$.

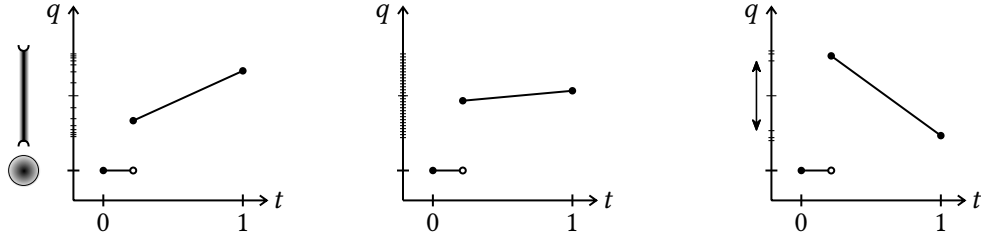


Figure 23: Three automorphism-equivalent trajectories in U , ticks and arrow in q representing the automorphism through scale and reflection, respectively

a given graphical pattern stands in for a stage; constant, zigzag, slope, or instantaneous transition. As I have already alluded to, in Croft's system the first three types of trajectories cannot be chained together unless separated by an instantaneous transition. Let us define a **SUBTRAJECTORY** as a trajectory from a certain subinterval of the time window, coinciding with the canonical trajectory. I argue explicitly that not any subtrajectory should be eligible as a profiled stage; rather, only those whose domain is connected and which are homogeneous in some sense should qualify.

Definition 3.3. If $f : B \rightarrow C$ is a function and B' a subset of B , then one can always form a less complete function that can only take elements of B' as antecedents: that function is called the **restriction of f to B'** , defined by $(\forall b \in B') f|_{B'}(b) = f(b)$ and notated $f|_{B'} : B' \rightarrow C$.

If $f = p : I \rightarrow S$ is a canonical trajectory, call any restriction of p a **subtrajectory of p** .

I propose to account for this by defining stages using **HOMOGENEOUS SUBSPACES** which partition the phase space, and transitions as points where the trajectory moves between homogeneous subspaces, thus giving a reason why the different graphical patterns must be separated. The partition into homogeneous subspaces in turn allows us to cut up the canonical trajectory into several **HOMOGENEOUS STAGES**, each encompassing one entire “stay” in some homogeneous subspace. Both homogeneous stages and instantaneous transitions that happen between homogeneous subspaces, as well as sequences thereof, are available for profiling.

Definition 3.4. Homogeneous subspaces are elements in a collection $(X_\alpha)_{\alpha \in A}$ of non-empty, pairwise disjoint subsets of the conceptual space.

Each of those X_α will be a connected subset encompassing all points of a certain “kind” within X . In particular, I assume, in keeping with the special status of the rest phase in Croft's system, that $\{g\}$ is its own homogeneous subspace. Though they may be reduced to a singleton, this is not always the case. Formally, a homogeneous state is defined in the following way:

Definition 3.5. Let X be a phase space and p be a trajectory within X . Given a homogeneous subspace X_α and a connected component J of the preimage $p^*(X_\alpha) \subseteq I$, call a **homogeneous stage of p at X_α** the subtrajectory $p|_J^{X_\alpha} : J \rightarrow X_\alpha$.^{ao}

^{ao}If $g : B' \rightarrow C$ and C' is a subset of C such that we have $(\forall b \in B') g(b) \in C'$, define the **corestriction of g to C'** as the function $g|^{C'} : B' \rightarrow C'$ whose codomain is now C' instead of C , and that satisfies $(\forall b \in B') g|^{C'}(b) = g(b)$. All else being equal, there are fewer functions into C' than into C , since there are fewer elements in C' to send elements of B' to.

Because the homogeneous subspaces form a partition of X , the collection of all stages of a trajectory describes the trajectory in full. Note that punctual homogeneous stages are not excluded, since the domain can be either an interval or a point.

The transitions between homogeneous subspaces that a trajectory follows can also be part of a trajectory's profile. Therefore, we need to define **TRANSITION POINTS**: time points none of whose neighborhoods are sent entirely inside one subspace. This means there are always two points arbitrarily close to such a point that are sent to different homogeneous subspaces.

Definition 3.6. A **transition point for p** is a point $t \in I$ such that $(\forall N \in \mathcal{N}_t) (\forall \alpha \in A) p_*(N) \not\subseteq X_\alpha$. There might indeed be two distinct homogeneous subsets such that p sends points immediately before t to X_{α_1} and points immediately after t to X_{α_2} , i.e.:

$$(\exists \alpha_1 \neq \alpha_2 \in A) (\exists \varepsilon > 0) (t - \varepsilon, t) \subseteq p^*(X_{\alpha_1}) \wedge (t, t + \varepsilon) \subseteq p^*(X_{\alpha_2}).$$

In that case, the tuple $\langle t, \alpha_1, \alpha_2 \rangle$ is called a **labeled transition point**.

Finally, I define a **PROFILED SUBTRAJECTORY** as a sequence of consecutive stages and transition points.

Definition 3.7. A **profiled subpart of the trajectory p** is an alternating list of homogeneous stages and labeled transition points of p such that the two consecutive elements of the list have abutting domains. This means that for any two consecutive elements, transition point is an infimum or supremum of the homogeneous stage's domain (which is always an interval).^{ap}

Aspectual types Now that profiling is properly formulated, homogeneous subspaces can be described for each of the phase spaces given in the preceding section: see Appendix B. We can finally define aspectual types formally:

Definition 3.8. Let X be a phase space with homogeneous subspaces $(X_\alpha)_{\alpha \in A}$, $p : I \rightarrow X$ be a trajectory within X with finitely many stages, and $\hat{p}$ be a profiled subpart of p (whose stages depend on the homogeneous subspaces). The tuple $\langle X, (X_\alpha)_{\alpha \in A}, p, \hat{p} \rangle$ defines an **aspectual type**.

Then for each aspectual type, the phase space and canonical trajectory are as described in Section 3.3, while the homogeneous subspaces are given in Appendix B and the profiled subpart is taken to be identical to Croft's, as displayed in Section 3.1.

Generalizations on canonical trajectories In addition to being continuous functions, it seems that several additional features are common to all canonical trajectories. For instance, they traverse all homogeneous subspaces of the phase space, and they only go through a finite number of stages, or equivalently have finitely many transition points. This seems a reasonable, parsimonious constraint, as it excludes the perfectly legal continuous function $I \rightarrow S$ depicted in Figure 30, that nevertheless sends an infinite, non-connected set of points to c . This kind of trajectory would only seem useful for examples like the following:

^{ap}This is different from a "pure" subtrajectory because we want to distinguish a homogeneous stage from the combination of that stage and an adjacent transition. But a transition point might also be part of the domain of an adjacent homogeneous stage, so that the subtrajectory corresponding to both would be identical to just the homogeneous stage; hence a list to keep record of the parts that are profiled.

- (24) The ping-pong ball bounced back and forth between the two planes, faster and faster as they came together, until it came to a stop.

However, this quite contrived example seems more parsimoniously described as a t -bounded activity instead of a new kind of trajectory. When the finiteness condition is enforced, the stages of a canonical trajectory can be numbered by order, and the profiled part of a trajectory can be described by giving a range of natural numbers specifying the profiled stages (and transition points). Furthermore, this forces any transition point to be surrounded by two extended homogeneous stages. This is not necessarily the case for general continuous functions, as shown by the previously mentioned trajectory of Figure 30: the accumulation point is a transition point, but does not have an extended homogeneous stage on its left.^{aq} Also, the profiled subpart of a trajectory may not include the initial stage, unless there is only one homogeneous subspace, as in the case of P .

Second, canonical trajectories are homogeneous in a neighborhood of the endpoints 0 and 1: the initial and final homogeneous stages always extend into the interval I some positive amount. Put differently, transition points are only allowed in the interior of the interval, within $(0, 1)$. For the initial stage, this happens automatically given that the preimage of $\{g\}$ is a neighborhood of 0 (since $p(0) = g$ and $\{g\}$ is a neighborhood of g), but the same cannot be said of the final stage. The canonical trajectory for accomplishments (Figure 17a) could just as well have a punctual final stage, for instance. However, complying with this mild constraint allows for a more uniform treatment of result states, on a par with those of directed achievements.

Third, homogeneous spaces should **respect automorphisms of the phase space**: if two points are sent to one another by automorphism, then they must belong to the same homogeneous subspace.^{ar} This means there are at most as many homogeneous spaces as automorphism classes of points in the space. (The only two cases with a discrepancy are the directed phase spaces I and D , for which the initial point $0_{I/D}$ is lumped together with the subsequent open interval.) This generalization is quite important: it ensures that stages indeed correspond between different trajectories in the same equivalence class. If automorphism classes of points were allowed to span several homogeneous spaces, different members of the equivalence class could feature a different sequence of stages, the homogeneous stage of one could even correspond to several stages and transitions for the other.

Fourth, all canonical trajectories, except those for point states and cyclic achievements, can be assumed not to feature any backtracking; this assumption makes them intuitively simpler, and reduces the number of candidates for canonical trajectories (e.g. excluding Figure 17c). More precisely, the generalization is that for any canonical trajectory, the preimage of any point in the phase space is connected: the trajectory might stay at a point for a while (extended interval), or simply pass through it during an instant (singleton), but not come back through it (which would give a non-connected preimage).

Point states and cyclic achievements The distinction I have drawn between homogeneous stages and transition points allows us to account for the difference in formalization between point/transitory states and the corresponding achievements. These are the two pairs of aspectual types where the only difference is in profiling.^{as} This is especially crucial for point states and cyclic achievements

^{aq}Indeed there is no predecessor to the stage at $2/3$ for this trajectory; the order between the stages is essentially that of the ordinal $\omega + 1$. By contrast, the trajectories depicted in Figures 16a and 16d have a transition point at $1/2$ which is indeed surrounded by two extended homogeneous stages.

^{ar}Two points $x, y \in X$ are in the same automorphism class if $(\exists \varphi : X \rightarrow X) \varphi \text{ automorphism} \wedge \varphi(x) = y$.

^{as}This might be attributed to two simultaneous causes. First, the plausible requirement that the non-profiled part of a trajectory be quite simple; for instance, that it have very few stages, maybe as few as two. Second, it seems that an inception stage into a non-singleton homogeneous space might as well be conceived of as the inception of a singleton state. This coincides with the analysis of the verb *start* and could be understood as a more abstract operator *START*; see

(which use the trajectory in Figure 16a), because they present a transition point which coincides with a homogeneous stage at the subspace $\{c\}$. Indeed this punctual homogeneous stage stands out as the only one in the aspectual classification. The associated transition point is neither the start of a new extended homogeneous stage nor the end of an old one; it makes up its own stage, lying in a different homogeneous subspace than all other points around it. This makes these points much less natural to express in the system. By dropping this exceptional, punctual homogeneous stage, we may have two extra generalizations. First, **all homogeneous stages would be extended**, providing a much neater distinction between homogeneous, extended stages and heterogeneous, punctual transitions. Second, we can make **all trajectories be non-backtracking**. While I leave the problem of point states unanswered for this thesis, I propose a different account of semelfactives/cyclic achievements in Section 3.6.

3.5. Directedness

In his system, Croft (2012) unearthed the notion of **directedness** (the ‘scalar change’ of Beavers 2008, 2011): a category that grouped together all aspectual types whose trajectory finished higher up on the q scale than it started. This includes essentially all aspectual types except states, cyclic achievements and undirected activities. Two overlapping subtypes exist: those with a definite result state, bringing together directed achievements and all types of accomplishments but excluding directed activities; and those that continually increase along the q dimension, including incremental accomplishments and directed activities and achievements, leaving out nonincremental accomplishments. Of course, accomplishments and achievements are distinguished by the extendedness of the change stage.

One problem with Croft’s characterization of this class is that it hinges on the relative heights of points on the q scale; but this disposition is purely conventional (Croft 2012, pp. 59–60). In this section, I formulate his condition in terms of properties of the phase spaces and trajectories within it, via an **advancement relation between points**. I should also like to suggest that nonincremental accomplishments are exceptional in this classification, and that they could in fact be done away with entirely. The only extended and telic aspectual type that remains is simply that of incremental accomplishments.

3.5.1. Advancement relation

I propose that the directedness property of Croft can be formulated in terms of trajectories: directed phase spaces are those with a definite notion of closeness to that rest phase. More precisely, in those spaces the points can be ordered by their closeness to the open point, using the fact that points closer up need to be crossed before those further away, while the converse is not true. Directed spaces are then exactly those for which this relation is a total order; and spaces with a result state are those in which one of the points is a greatest element for this order relation.

Definition 3.9. Let x, y be two points in a phase space X with open point g . Say that y is **more advanced than** x , and write $y \geq x$, if and only if $(\forall p : I \rightarrow X) [p(0) = g \wedge p(1) = y] \implies [(\exists t \in I) p(t) = x]$.

By design, the rest point g is always less advanced than any other point in the space. We can prove that the advancement relation is indeed a total order on the following spaces: S, I and D .^{at} In the former two spaces, it also has a greatest element: c in S and 1_I in I . In fact, removal of this greatest element in

Section 3.6.

^{at}For the latter two spaces, this uses the fact that $X \setminus \{x\}$ is disconnected for all $x \in \{0_{I/D}, 1_{I/D}\}$.



Figure 24: Candidate spaces for nonincremental accomplishments (the latter two taken from Figure 20)

either case yields another directed phase space: P and D respectively; in a sense, S and I are “natural” equivalents to P and D , with a greatest element added.^{au} Indeed, the space P also trivially satisfies this order condition. However, because permanent states and transitory states profile a constant stage instead of a transition or “active” stage, they can reasonably be excluded from the class of directed aspectual types.

Definition 3.10. An aspectual type is called **directed** iff the advancement relation is a total order on its phase space, and the profiled stage of its canonical trajectory is non-constant.

On the other hand, the space U , used for undirected activities, only has the point g as smallest element for the advancement relation, while no other non-trivial relations hold. Indeed, for any point $y \in U \setminus \{g\}$ there is always a trajectory that only traverses g and y , so that there is no other point that is less advanced than y (these trajectories are similar to that of Figure 16d). Therefore, the advancement relation is not a total order on U , and the aspectual type of undirected activities is not directed.

3.5.2. Nonincremental accomplishments

The previous definition formally reproduces Croft’s classification, save for nonincremental accomplishments. I know of no phase spaces that could plausibly be used for this aspectual type, satisfy the conditions stated in Section 3.3.6, and for which the advancement relation is indeed a total order. It would be most natural for this space to be a “result-state addition” to the space for undirected activities U , so that removal of the result state yields back the corresponding activity type just like in Croft’s (2012) system. This could yield at least two spaces: one where the result state is an open point (Figure 24a), and one where it is a closed point (Figure 24b); in both cases, the new point is accessible from any point in the remainder of the active subset. However, even in those spaces the advancement relation would not be a total order, since this is not even the case on the subset U . These spaces would also break the patterns of previously mentioned phase spaces: for instance, neither’s active subset is a manifold, and the first one even has more than one open point.

The same goes even for those spaces whose advancement relation (relative to a chosen open point) is indeed a total order: those are featured in Figures 24c and 24d. Outside of the system presented thus far, these spaces are conceivable, and could nicely explain the alternation between two different states ascribed by Croft to undirected activities. However, they too break the pattern of previous phase spaces (having more than one open point), and would require change in other parts of the system if we wish to keep the analogy between activities and accomplishments.

In any case, it is clear that nonincremental accomplishments fit much less neatly into this formal version of the system. A possible solution, which I pursue here, is simply to **deny the existence of**

^{au} More precisely, for $X \in \{P, D\}$, there is a unique space $X^\top$ (up to isomorphism) such that (a) the advancement relation $\geq_\top$ is an order on $X^\top$, (b) there is an element $\top \in X^\top$ which is a greatest element for $\geq_\top$ (i.e. $(\forall x \in X^\top) \top \geq_\top x$), (c) the subspace $X^\top \setminus \{\top\}$ is isomorphic to X .

a separate class of nonincremental accomplishments. The argument raised by Croft (2012) to isolate this class is that it does not combine readily with measure adverbials:

(25) *She has fallen asleep a quarter of the way.

Other predicates are taken to behave in the same way because they do not usually feature measurable progress. Rothstein (2004, pp. 98–99) used them against a homomorphism analysis of telicity, because the extent of the event is not determined by the extent of the overt theme.

- (26) a. Harry was repairing the computer.
b. John is painting a portrait of his cat.
c. Leave me alone—I’m solving Rubik’s cube. (Croft 2012, p. 41)

This class could in fact include many more construals of events usually conceived of as incremental accomplishments. For instance, *writing a book* can be nonincremental just like *painting a portrait*, if occasional edits or major revisions keep happening all over the manuscript; and the same can be said for *reading a textbook thoroughly*.

If we take aspect to pertain to the qualitative, and not the quantitative unfolding of events, then the inability of certain predicates to occur with measure adverbials should not be in any way relevant to their aspectual classification. Because of this, **I do not regard objective measurability of progress as an aspectual property**. This is clearly seen in my use of self-equivalence of the phase space, which discards any absolute “measure” of the event in favor of relative measures, in particular with respect to the terminus. This choice goes hand in hand with a view of the *q* dimension as reflecting aspectual meaning, instead of event-type meaning. Thus, the linear continuum is simply a given of our conceptual capacity, not a scale on which the entity can be placed at any given time point provided one has a snapshot. Instead, the event is conceived as proceeding along it, whether or not it may involve occasional dead ends.

In fact, though quantitative measure is disallowed with some achievements, presumably based on world knowledge, “nonincremental” accomplishments definitely allow qualitative measure, as suggests their use with *almost*. As Dowty (1979, p. 58) notes, *almost* can always denote near-occurrence of an eventuality; however, accomplishments (unlike activities and achievements) license an additional near-completion reading, so that sentences containing them are ambiguous. I hold that nonincremental accomplishments feature the same ambiguity as that of incremental accomplishments:

- (27) a. Harry almost repaired the computer.
b. Fermat had almost proven the theorem, but lost the notes before he could complete it. (Near-completion reading)

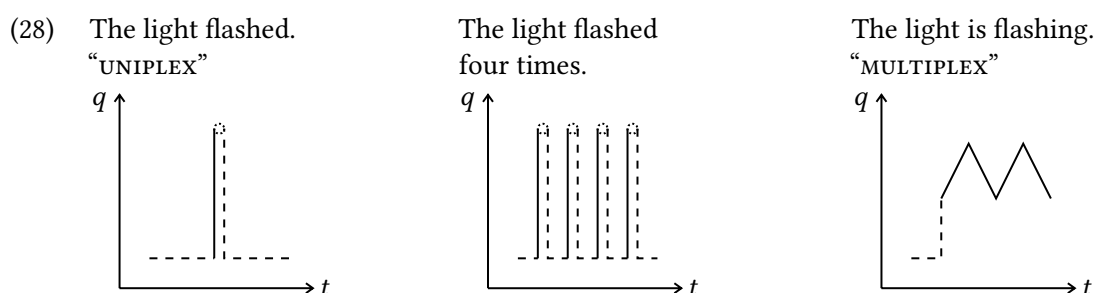
The two readings here are one where Harry almost got to the repairing activity but never did, and another where Harry did look for a solution to the computer’s malfunction, and got close but did not actually repair it in the end (whether the near-completion was a red herring or the project was abandoned). This should not be possible if no directed scale was present underlyingly to compare different subevents with respect to the ideal result state. Therefore, **I analyze Croft’s (2012) nonincremental accomplishments as regular, incremental accomplishments**.

3.6. Changes of granularity and quotients

3.6.1. Croft's operations

Croft presents several mechanisms of aspectual construal that explain many predicates' potential to be construed under several aspectual types. One of those, **selection**, involves picking another profiled stage in the same canonical trajectory (see Example 4 on p. 8), while keeping the same phase space. Few constraints are naturally imposed on this operation in the current version of my system, since no stage is favored over any other. That being said, it is conceivable that extraneous constraints hold for this mechanism, like that the shift in profiling may only happen between adjacent stages.

Two other cognitive operations are taken to play an important role in aspectual potential: **structural schematization** and **scalar adjustment**. The former can construe repeated or back and forth events as undirected activities by considering disjoint profiled segments as a continuous one, analogous to the shift between simplex *hairs* and multiplex *hair*:



The latter can adjust the time scale to toggle between instantaneous and extended transitions, or the qualitative scale to make q states appear or disappear:

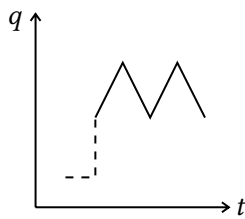


However, it is not always obvious whether to interpret a given aspectual alternation as belonging to one operation or the other, largely because the way these operations work is left implicit. For instance, the difference in (28) between iterated cyclic achievements and undirected activity could also be analyzed as a change of granularity, where the intermediate state between flashes is seen as somehow pertaining to the flashing event. In fact, this is exactly how *opening and closing the window* is analyzed in Russian (p. 122). Conversely, the shift in (29) between undirected activity (behavior) and permanent state (disposition) could be interpreted not as a change in granularity as Croft proposes, but as a structural schematization that construes repeated behavior as manifesting an essential disposition of the individual.

Of the two, scalar adjustment is somewhat more prominent in the system, with multiple examples given. The same alternation as in (29) is also ascribed to the contrast in (30); though for the former the usual construal of the predicate *be polite* is the coarse-grained permanent state, and for the latter *hike* is most usually a fine-grained undirected activity.

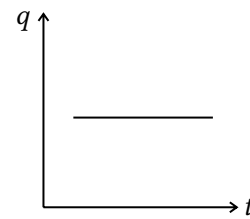
- (30) She is hiking through the woods.

FINE-GRAINED



She hikes every weekend.

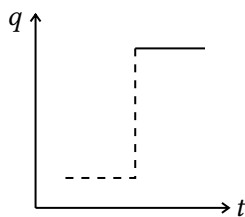
COARSE-GRAINED



A similar process happens in the shift illustrated in (31), simply changing the usual activity in the Progressive to a transitory state characteristic of inactive actions.

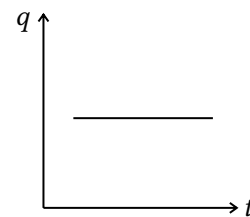
- (31) He is lying on the bed.

FINE-GRAINED



The forest lies to the east.

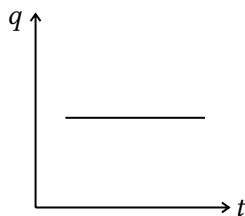
COARSE-GRAINED



Note that this contrasts with the construal of certain states in the Progressive, which Croft suggests could be an instance of a coarse-grained view in both dimensions, thus revealing extra points for the entity to traverse in a directed way:

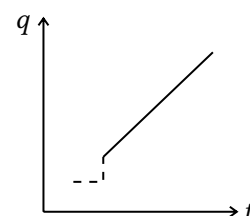
- (32) She resembles her mother.

FINE-GRAINED



She is resembling her mother more and more every year.

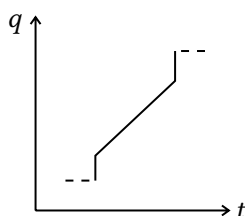
COARSE-GRAINED



The final kind of construal that Croft explains is the shift between fine-grained accomplishments (incremental or not) and coarse-grained achievements, showcased in the introduction and repeated here as (33).

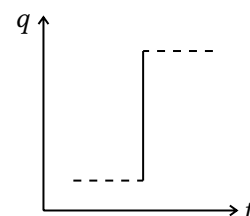
- (33) The bridge is collapsing.

FINE-GRAINED



The bridge collapsed at 9:15.

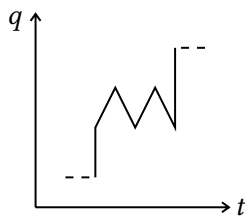
COARSE-GRAINED



A similar change is ascribed to the contrast in (34), swapping the incremental accomplishment for a nonincremental one.

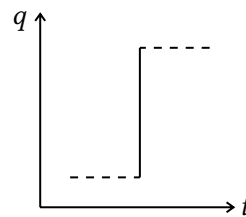
- (34) He died in three hours.

FINE-GRAINED



Jerry Garcia died on August 9, 1995.

COARSE-GRAINED



Naturally, the converse change between achievement and accomplishment, as in (35), pertains to the reverse kind of scalar adjustment; the only difference being that this predicate is usually an achievement instead of an accomplishment.

- (35) a. He reached the top in twenty minutes.
b. He reached the top at 9:15.

To make the two operations of structural schematization and scalar adjustment formal, I propose and to account for them at the level of phase spaces, by seeing the operation as a certain kind of **continuous functions** between phase spaces, which only secondarily affects trajectories. Then, these continuous maps change both the phase space and the canonical trajectory, yielding another aspectual type as a result. In suggesting this formal device, I clean up the different operations by bringing them all under the banner of *quantitative* scalar adjustment: both temporal scalar adjustment and structural schematization can be analyzed using this formal version of qualitative scalar adjustment, making them superfluous. The continuous functions between qualitative phase spaces, called PHASE TRANSFORMATIONS are taken to reflect aspectual potential for conversion between different spaces, just like Croft suggests. I propose three kinds of qualitative scalar adjustment and discuss each in turn.

3.6.2. Full collapse into a point

Permanent state–activity alternations What is common to examples (29)–(32) is that the Present leads to a much simpler trajectory, that of a permanent state, disregarding the contour of the fine-grained trajectory. This happens whether or not the predicate is usually construed as an activity or as a permanent state. I propose to account for this via simple continuous functions between phase spaces, called PHASE TRANSFORMATIONS. In the cases at hand, the phase transformations relate the aspectual types of transitory states, undirected activities and directed activities on the one hand, and that of permanent states on the other. Because the transformations relate phase spaces to one another, this places the focus away from temporal manipulation of the trajectory and onto **manipulation of the qualitative space**. Indeed, I assume no kind of temporal scalar adjustment, and place the burden of aspectual potential entirely on the conceptual spaces and transformations between them.

Redefining granularity In doing so, I take a different view than Croft on certain states in the Progressive, such as (32). Instead of considering the directed activity construal as a coarse-grained counterpart of the permanent state, I view it as **fine-grained** because it involves **more distinctions in the qualitative space**. More generally, I define a COARSE-GRAINED SPACE in relative terms, as one that has a more cohesive set of points, and possibly fewer distinctions between its points. Formally, this means that it sits as **the target space of a phase transformation**: any continuous function can only add cohesion to an initial space, even collapsing some points together as it transitions into the target space; the reverse situation is disallowed.^{av} This different definition explains the discrepancy in analysis with

^{av}If only because no source point can be split by a function: any function has a unique target image on any source point.

Croft, who uses ‘coarse-grainedness’ to refer to a greater intuitive “distance of observation” to the q scale. Though this can be a useful analogy, it leaves some ambiguity: a closer-up, fine-grained view on the y axis does magnify differences along it, but a closer-up view on the x axis reduces the perceived rate of change along y , which therefore seems *less* fine-grained. In the case of *resembling her mother more and more*, one could see an instance of a coarse-grained x -axis view, because the undirected activity involves a wider timespan. By contrast, a simultaneous zoom on both axes will smooth out any curvature that a trajectory has, but crucially not remove its slope – this is what happens for the Earth’s curvature.

By rejecting temporal manipulation and focusing on qualitative transformations, I hope to illuminate what is meant by granularity, and simplify the relationship between different aspectual types. In this new understanding of granularity, it is now much more economical to describe the undirected activity in (32) as a fine-grained construal. This simultaneously unifies these special cases of the Present as coarse-grained; and clarifies the Progressive by contrast, which usually leads to more fine-grained but homogeneous construals.

Collapse transformations The continuous functions that I assume in the kind of alternation displayed in (29)–(32) have the peculiarity of being **partial functions**: they are defined everywhere except at the rest point, which is simply discarded. This seems to be the simplest way to account for the fact that the rest phase does not matter anymore, by forgetting it entirely, although it requires expanding slightly what is meant by ‘function from X to Y .’ Hence, I notate the type of these special functions, which are restricted to the active subset of the domain, using an arrow with an initial loop instead of the usual, undecorated arrow: $X \looparrowright Y$ instead of $X \rightarrow Y$. However, this move seems warranted inasmuch as the construals in all these examples profile a **homogeneous stage** of the canonical trajectory within the active subset. In contrast to transitory stages, this special, atelic profile clearly backgrounds the rest phase.

Because the coarse-grained trajectory in each example is that of a permanent state, occurring within the point space P , the partial functions will all have P as their target. As a result, all these partial functions are simple to describe, because there is only one choice for where to send any non-rest point: they are represented in Figure 25 and written as functions of the form ‘ $y \mapsto x$ ’ where y is a free variable denoting any point in the active subset of the source space. These functions can be thought of as **full collapses** of the phase space into a point. Indeed, when the canonical trajectory for an aspectual type is composed with this collapse transformation, it becomes the only trajectory within P , the constant trajectory.

Recipe for aspectual alternations The procedure to change aspectual type is the following.

Definition 3.11. Let T be an aspectual type with phase space X and canonical trajectory p , such that the profiled part $\hat{p}$ of p is a homogeneous stage. Assume that π_X is a collapse transformation of X into P . From these two pieces of data, one can form a new aspectual type $T^\#$, called the **full collapse of T** , by defining it to have:

- P as its phase space;
- $\pi_X \circ p : I \rightarrow P$ as its canonical trajectory;
- $\pi_X \circ \hat{p}$ as the profiled stage of this trajectory.

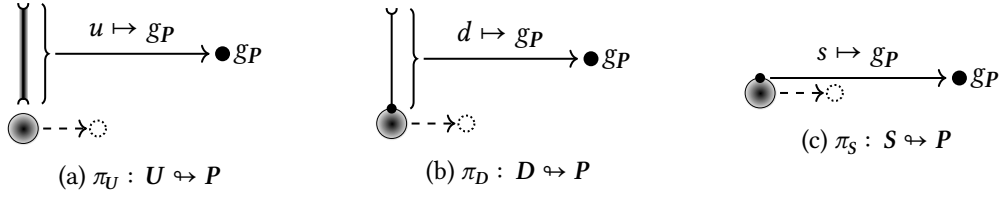


Figure 25: Collapse functions from three phase spaces into the one-point space P

In each case the phase transformation itself has a specified direction, which makes it very easy, given a trajectory within its source space, to compute the corresponding trajectory in its target space. This does not force the aspectual type of the source space to be the default one, however. Construal alternation could be a sort of inverse problem: given a coarse-grained trajectory in the target space, find an appropriate, more fine-grained one that reduces to it when using the phase transformation.

Note that in principle, because π_X is not a total function, the composition $\pi_X \circ p$ is also partially defined: it is only defined on the preimage of the active subset, i.e. $p^*(X \setminus \{g\})$. This can be remedied by simply forgetting about the undefined points, and relabeling the remaining points back into a full interval I . This is easily done if the rest stage $p^*(\{g\})$ is an initial subinterval of I , which is the case for all canonical trajectories. Otherwise, this procedure could still be used, with precautions, to glue the active stages of a non-canonical trajectory together back into an interval. In this way, one can easily “forget” about intermediate rest stages in the case of iterated events. I do not pursue this further, as all the trajectories I consider in this thesis are canonical trajectories.

Fuller account of the Present Note that by contrast to the atelic aspectual types mentioned, I do not consider telic predicates in the Present as part of this more general cognitive operation of scalar adjustment. But of course, the habitual use of the Present is routinely available both for atelic and telic predicates – though this is oddly not present in Croft’s (2012) discussion of the Present (pp. 149–152). In that respect, (36a) is hardly different in from (36b). For that matter, the full collapse transformations proposed here are also conceivable for the space of accomplishments I .

- (36) a. Caster runs 800 m in two minutes every year.
b. She hikes every weekend.

While analyzing (36a) with a permanent state construal is a possibility entertained by Croft, it is also quite plausible that a more precise, naturally unified account of habitual readings is lacking. I leave this question for future research.

3.6.3. Partial collapses

Another effect of scalar adjustment for Croft is the two-way potential for construal between accomplishments and achievements, as shown in (33) and (34). The reverse shift into finer granularity, as in (35), is commonly permitted for “achievement predicates,” mostly in the Progressive and with Container adverbials. This is true for achievements that more easily expand into incremental accomplishments (such as *reach*), or those that usually become nonincremental accomplishments (*fall asleep*, *leave*). As I signaled in Section 3.5.2 however, I choose the more economical option of only having incremental accomplishments in my typology and to group the nonincremental accomplishments of other accounts inside this class. Therefore, there is only one phenomenon to account for here, although the construal can go either way depending on the default construal of the predicate.

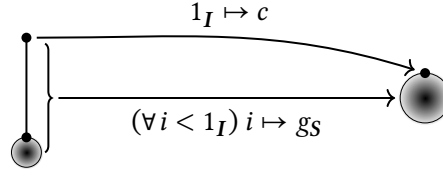


Figure 26: The quotient map $\sigma_I^S : I \rightarrow S$

Partial collapse as a quotient map For this kind of scalar adjustment, the coarse-grained alternative clearly retains the transition of the fine-grained accomplishment, so the space P will not do. For this reason, I use the less radical **quotient maps** as phase transformations. Quotient maps are defined by a domain, and which points of that domain they collapse together. The codomain is then determined from these data: each point of the codomain corresponds to a set of points of the domain that have been collapsed together, and no other points are present in the codomain; the topology of the codomain then follows from the topology of the domain, and the quotient map is continuous under this topology. In this sense, quotient spaces really provide a coarser version of the original space, since they have fewer points. In the limiting case, all points are collapsed together and the codomain is the one-point space P .

For the usual granularity alternation between accomplishments and achievements, the domain of the quotient is I , the closed interval with rest point. Within this space, I propose to collapse the active, non-result subset into the rest phase, while keeping the terminus phase as its own distinct point in the codomain: this quotient map σ_I^S (henceforth σ for short) is depicted in Figure 26, along with its action on the elements of I . The resulting codomain is simply the Sierpiński space S : then, the partial collapse σ of I into S is a continuous function. It is also a total function, by contrast to the full collapse transformations of the previous section which were not defined on the rest point.

Relation between aspectual types This partial collapse naturally relates the phase spaces of accomplishments and achievements, and even the two aspectual types. At the level of aspectual types, σ acts in the following way: regarding the phase space, that of accomplishments I becomes that of achievements S . The canonical trajectory of accomplishments, when composed with this quotient, becomes exactly that of achievements, up to reparametrization (Figure 27a). Finally, the profiled part of the canonical trajectory becomes a part of the new trajectory which cannot be profiled: first, the first “transition” stage is not a transition stage anymore; second, the homogeneous stage is not actually complete, since it is missing an initial segment. Both of these illicit stages are discarded by the adjustment operation Γ , leaving just the transition stage as the profiled subpart (Figure 27b). Thus, the profiled subpart is trimmed so as to keep only the legal transition phase between $\{g\}$ and $\{c\}$.

Definition 3.12. Let p be a trajectory in a space X with homogeneous subsets $(X_\alpha)_{\alpha \in A}$ and $\hat{p} = \langle \hat{p}_0, \dots, \hat{p}_n \rangle$ be a candidate profiled subpart of that trajectory. The **trimming operation** Γ transforms $\hat{p}$ into a legal profiled subpart. It does this by only keeping legitimate stages as defined in Section 3.4.2:

- labeled transition points $t \in I$ such that ;
- a sublist of contiguous homogeneous stages $\hat{p}_i : J_i \rightarrow X_\alpha$ has a corresponding element in $\Gamma(\hat{p})$ only if its collation $\bigcup_i \hat{p}_i : \bigcup_i J_i \rightarrow X_\alpha$ is indeed a homogeneous stage, i.e. such that $\bigcup_i J_i$ is a connected component of $p^*(X_\alpha)$. In that case, only the collation $\bigcup_i \hat{p}_i$ is kept as an element in the list $\Gamma(\hat{p})$.

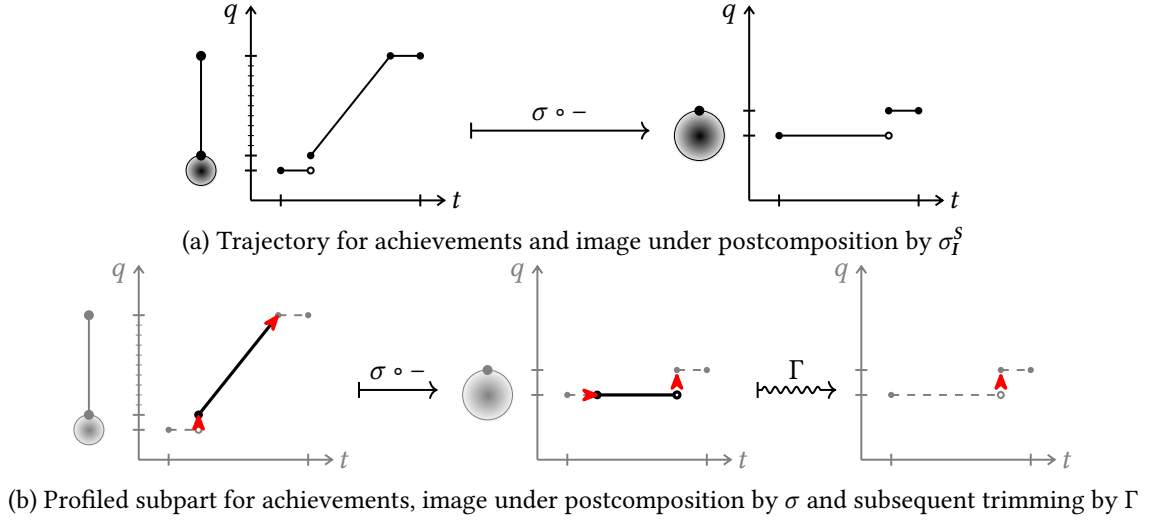


Figure 27: Action of the quotient on the phase space, trajectory and profiled subpart

Therefore, the two aspectual types are related via the action of σ , function of type $I \rightarrow S$:

- the phase space of accomplishments I becomes the codomain of σ , to wit the phase space S of achievements;
- the canonical trajectory of accomplishments $p_I : I \rightarrow I$ becomes $\sigma \circ p_I : I \rightarrow S$, coinciding with the canonical trajectory of achievements;
- the profiled subpart $\widehat{p}_I$ becomes $\Gamma(\sigma \circ \widehat{p}_I)$, coinciding with the profiled subpart for achievements.

This quotient transformation, and the more general trimming operation Γ , therefore predict the special relationship between accomplishments and achievements, which are very easily converted one into the other. Thanks to it, we now have a precise notion of the alternation between those two aspectual types, and why the q scale for accomplishments can be regarded as fine-grained against that, coarse-grained, of achievements: it is simply the domain of a map that collapses distinctions between points.

Final stage It is notable that only the final stage is kept in the transformation between accomplishments and achievements. This comes from the fact that I assume the result state to be identical between two construals of an event as an accomplishment and as an achievement. I cannot motivate it further, however: even under an achievement construal with a time adverbial, it seems that the entire duration of the event should be included within the adverbial. Say that Kim started running an ultramarathon on 30 August, 2024 and passed the finish line a few minutes after midnight on 1 September. In this case, (37a) is much better than (37b) which only includes the endpoint. Nevertheless, (37b) is still more acceptable than (37c), and this fact could comfort my analysis somewhat. I lack the scholarship on time adverbials to expand on this further.

- (37) a. Kim ran an ultramarathon between 30 August and 1 September, 2024.
 b. ?Kim ran an ultramarathon on 1 September, 2024.
 c. ??Kim ran an ultramarathon on 30 August, 2024.



(a) The partial function $\tau_I^S : I \multimap S$, relevant for FINISH (b) The contraction $\xi : I \multimap S$, relevant for START

Figure 28

Aspectual verbs Given that the standard conversion between accomplishments and achievements only keeps the equivalent of the final stage in the profile, it is natural to ask how this construal operation might differ from that forced by the aspectual operator FINISH. For this, I propose that a distinction can be drawn between the partial collapse $\sigma : I \rightarrow S$ and another, partial, phase transformation $\tau : I \multimap S$. I define τ_I^S (τ for short) as having the same effect as σ on all points, except that it is undefined on the rest phase. This transformation is represented in Figure 28a. Analogous collapsing-forgetting operations, keeping a non-trivial space in the codomain, cannot easily be defined on other phase spaces: those only have two homogeneous subspaces, one of which is the rest phase. Thus, the only phase transformation into another phase space that forgets the open point is a full collapse from Section 3.6.2. Further classification of phase transformations is needed to put this possibility in perspective.

A clearly analogous phase transformation could be studied for the operator START as well: I call it $\xi : I \multimap S$. Instead of being undefined on the rest phase, ξ forgets about the result phase and collapses the other active points into a single result phase (hence the new symbol $\multimap$). Delineating properties of this transformation would help determine why *start* does not readily combine with achievements, although it does with all extended aspectual types: accomplishments, activities, and transitory states.

Two remaining aspectual verbs that may require a different approach are STOP and CONTINUE. For STOP, some transformation should be available that allows instantaneous jumps into the rest phase even from arbitrary points of the space: while this property holds for U , it does not for either D or I . Furthermore, if the transformation was assumed analogous to the ones presented so far, it would need to relate extended events to achievements, so that the result phase of the achievement (a closed point) somehow corresponds to the rest phase of the extended event (an open point). This is no easy feat, since open points are naturally sent to open points by continuous functions. Regarding CONTINUE, this would seemingly require separating the extended stage of the event in two, one on either side of the “continuation point.” I do not know how to represent this, since I have taken phase transformations to respect homogeneous spaces. Although these aspectual operators are much more easily described in Croft’s system, they could suggest new avenues to change the formal system of the present, more regimented proposal.

3.6.4. Semelfactives as atomic minimal events

Croft analyzed semelfactives as cyclic achievements, inspired by Smith (1997): he notes that they are like achievements in being punctual but unlike achievements in that they are not predicates of change. His analysis reflects this, by tracing out a punctual change on the q scale that instantly reverts back, so that the entire event does not involve any change. One of the more puzzling features of semelfactives is their ability to be construed as activities when employed in durative constructions, with a multiple-event reading. Smith (1997, p. 30) suggested that the single and multiple versions of semelfactive events could be equally basic: blinking, coughing and knocking often occur in sequences. This motivates the use of structural schematization in Croft’s system, an operation which is also available for entities.

However, as I noted in Section 3.4.2, transposing Croft’s analysis of semelfactives as cyclic achieve-

ments into my system would imply the presence of a trajectory whose transition stage has the same starting and destination subsets in the space S . While conceivable, this would disrupt the otherwise uniform treatment of aspectual types; disregarding cyclic achievements as a separate class would allow for a neater classification, where all homogeneous stages are extended and no canonical trajectory backtracks. In addition, nothing in the phase space could force S to only feature essentially cyclic trajectories: bump-like trajectories (Figure 16d) will necessarily exist, too. To defend this rejection of cyclic achievements, I propose to analyze semelfactives as undirected activities, with a related, telic accomplishment construal that make counting single events possible. The activity construal is agnostic as to the number of minimal events involved, while the accomplishment construal by default only involves one. This accounts for the ambiguously telic or atelic behavior of semelfactives depending on construction.

Telic semelfactives The telic behavior of semelfactives on the single-event reading was pointed out by Rothstein's (2008). Indeed, contrary to earlier positions such as Smith (1997), Rothstein defended that they can be used in the Progressive to refer to a preliminary interval, making them subject to the imperfective paradox, when a single minimal event is interrupted:

- (38) a. John was knocking hard when he saw me, so he turned it into a tap instead (and didn't knock hard).
 b. Bill was kicking him when he saw me, so he stopped midway (and didn't kick him).

They also occur with the Container adverbial in *TimeInterval*. In the context of a pole vault, the single event reading is acceptable, measuring the whole temporal extent of the event from beginning to end:

- (39) John jumped in three seconds.

As was noted by Smith (1997, p. 46), this construction can also refer to the preparatory phase of a single event, which she calls 'ingressive':

- (40) a. I peppered Naima's nose, and she sneezed in five seconds.
 b. She slowly tapped his shoulder.

In this, semelfactives are hardly different from directed achievements.

She further remarked that these events do have structure, in contrast to canonical achievements like *reach*. Indeed, they involve specific trajectories, which she takes to be different from instantaneous change and necessarily extended. This differs from Croft's account, which readily allows a spike-like trajectory involving two-way change (cf. Figure 13d). Though it is conceivable for evidently extended events to be conceived of as instantaneous, I wish to vindicate this idea of extended semelfactives under an aspectual-type view of the qualitative dimension. Though construal as an achievement seems to be always available alongside construal as an accomplishment, I contend that it is more informative to view semelfactives as extended. This is also Rothstein's point in viewing semelfactives as activities with **naturally atomic minimal events**. However, Rothstein still firmly views semelfactives as activities. Her interpretation for their being telic is that she defines telicity as the property of denoting a set of atomic entities. In particular, "while telicity is not synonymous with being a predicate of change, predicates of change are naturally telic predicates. This is because changes are atomic, and the beginnings and endpoints of events of change are determined by the beginning and end of the change." If we do not take aspectual types to be defined by particular changes, however, this allows to view all

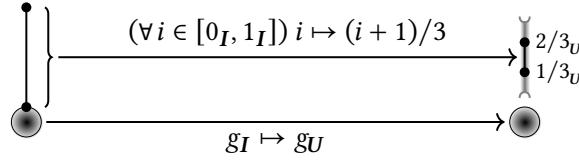


Figure 29: $\rho : I \rightarrow U$

atomic events squarely in the class of telic predicates: though semelfactives do not profile any overall change in the entity, they nevertheless take time to unfold and have a natural terminus, making them canonical accomplishments.

A phase transformation to pick out a minimal event I propose to reflect the alternation between accomplishments and activities using the phase transformation ρ given in Figure 29. This transformation sends the rest point of I to the rest point of U , while embedding the closed interval of I comfortably inside the open interval of U . Though I represent ρ as sending 0_I to $1/3_U$ and 1_I to $2/3_U$, this transformation is equivalent to any number of others, under self-equivalence of U ; I merely choose this particular map as a matter of convenience. Also, note that ρ is indeed continuous at 0_I , even though $\rho(0_I) = 1/3_U$ is drawn apart from $\rho(g_I) = g_U$. This is simply a necessary feature of the representation, but by definition the open point in U is in fact arbitrarily close to any point in the open interval, as the halo around the latter is meant to suggest.^{aw}

The fact that U contains “more points” than I could also seem problematic if the activity reading is more coarse-grained than the single-event reading. However, the image of I by ρ has a clearly coarser topology than I itself: it does not collapse any points, but it does erase the topological differences between the initial, middle and final points. The rest point now sits infinitely close to all of those points, and all their neighborhood systems are now isomorphic. In this respect, we again make more precise the shift into coarse-grainedness by which Croft analyzed the construal of repeated cyclic achievements as undirected activities, as in (28). Note that, by contrast to the alternation between Progressive and Present, here the Progressive can entail a coarse-grained view.

When considered in isolation, the space for undirected activities U has an entirely uniform active subset. In Rothstein (2008) and Dowty’s (1979) view, all activities have minimal events, which in my system is any trajectory that makes some non-zero progress along the open interval (as in Figure 19), as against trajectories that simply stay put at a point of the active subset of U . But when U is considered as the codomain of ρ , the transformation picks out a certain subpart of it, delimiting the extent that a **minimal event** should traverse in order to count as such; just like the trajectory for an accomplishment must traverse the space I to count as culminating. This partly vindicates Rothstein’s (2008) view of semelfactives as those activities whose minimal events are **naturally atomic**: in my system, those predicates that are also equipped with ρ are the semelfactive ones. When a minimal event is picked out, the rest of the open interval is then available for a trajectory to cross and represent the activity.

Dual interpretation of semelfactives These two spaces help explain the ambiguity of semelfactives in the Progressive: for a single-event reading, they behave as other telic predicates in referring the preparatory phase; in a multiple-event reading, they refer to the ongoing process. A remaining puzzle is the fact that semelfactives do not easily occur in the single-event reading with aspectual verbs like *start* and *finish*. An explanation for this fact is left to future research.

^{aw}More formally, a neighborhood basis of any point $u \in U \setminus \{g_U\}$ is given by $(u - \varepsilon, u + \varepsilon) \cup \{g_U\}$, ($\varepsilon > 0$), and indeed the preimage by ρ of any such neighborhood is itself a neighborhood of 0_I .

Base (IPFV)	Meaning	Derived (PFV)	Meaning	Semel.?	Source
<i>kášljat</i>	‘cough’	<i>kášljanut</i>	‘cough once’	✓	(Smith 1997, p. 246)
<i>stúcat</i>	‘knock’	<i>stúknut</i>	‘knock once’	✓	(Smith 1997, pp. 233, 246)
<i>dút</i>	‘blow (of wind)’	<i>dunút</i>	‘blow once (of wind)’	?	Wiktionary
<i>lí(va)t</i>	‘pour’	<i>liv(a)nút</i>	‘pour’	✗	Wiktionary
<i>usmexátšja</i>	‘grin’	<i>usmexnúšja</i>	‘start grinning’	✗	Wiktionary
<i>skolžíť</i>	‘slide’	<i>skolznút</i>	‘slide’	✗	Wiktionary

Table 1: Russian base and derived verbs with “semelfactive” suffix *-nut*

Base	Meaning	Derived (FREQ)	Meaning	Semel.?	Source
<i>välähtää</i>	‘flash once’	<i>välähdellä</i>	‘flash repeatedly’	✓	(Carlson 1981, p. 43)
<i>aivastaa</i>	‘sneeze once’	<i>aivastella</i>	‘sneeze repeatedly’	✓	Wiktionary
<i>nostaa</i>	‘lift’	<i>nostella</i>	‘lift (e.g. weights)’	✗	Wiktionary
<i>ostaa</i>	‘buy’	<i>ostella</i>	‘shop’	✗	Wiktionary
<i>muistaa</i>	‘remember’	<i>muistella</i>	‘reminisce, commemorate’	✗	Wiktionary

Table 2: Finnish base and derived verbs with frequentative suffix *-ella*

Profiling problems I have yet to explain how the shift from an accomplishment to an activity via ρ happens. The image of the canonical trajectory for accomplishments under postcomposition by ρ is indeed a canonical trajectory for an undirected activity. However, the resulting profile does not correspond to the right profile: under simple composition, the new candidate profile is not well-formed, but the trimming operation Γ does not fix it either. The candidate profile only highlights part of the full homogeneous stage of the new trajectory, and the only bona-fide part of the profile that remains is the initial transition from rest point to activity. Instead, if the resulting trajectory is meant to be an activity trajectory, we should expect its profile to be the homogeneous stage and nothing more. For now, I must ascribe this discrepancy to a shift in profiling; a better understanding of which profilings are allowed for which trajectory could help explain why this change is required.

Unexceptional semelfactives There is no assumption that either construal should be basic; and it could be the case that the class of predicates with such alternations could be different from language to language, based on the same formal mechanism. Preliminary investigations of verbs in Russian and Finnish suggest this is the case. Note that this is only a secondary source, which does not consist in native judgments or translations but in definitions on Wiktionary pages in English and in the source language; further confirmation for those facts is needed. In Russian, the function of *-nut* is to form perfective semelfactive verbs from the corresponding imperfective activity verbs: *kášljat* (IPFV) vs. *kášljanut* (PFV) ‘cough,’ etc. (Smith 1997, p. 246). However, other lexical items are involved in this alternation which are not usually taken to be semelfactives: see Table 1. Examining the kinds of events involved in this alternation, it seems reasonable to assume that other semantic properties are associated to this suffix, such as rapid or violent actions, which may occasionally get specialized for a semelfactive reading. However, without a more sophisticated aspectual analysis it does not seem warranted to ascribe semelfactives to an entirely different subclass of events. Conversely, Finnish distinguishes single events from multiple events using the frequentative suffix *-ella*, as in *välähtää* ‘flash once’ vs. *välähdellä* ‘flash repeatedly’ (Carlson 1981, p. 43). Analogous to Russian, several non-semelfactive telic predicates are subject to this alternation, displayed in Table 2. Based on this data, there is no particular reason to distinguish single-event semelfactives from other telic predicates.

Indeed, it could be defended that semelfactives are simply accomplishments that do not need any

modifier, whether incremental theme or conventionalized degree, to behave as such. This is a special fact, when most accomplishments need a definite object for the single-event reading to come through. Under this perspective, the single-event reading of *sneeze* could be to its multiple-event reading as *eat an apple* is to *eat apples*. This analogy could be complemented with directed predicates of change which also allow an accomplishment reading when a specific degree of change is contextually present (*grow up*) and a directed activity reading (*grow*). Unlike these other predicates, they **naturally** denote an atomic, extended event. Other lexical items that denote atomic events, like *reach*, are not extended; only “resettable achievements” like *fall* seem to share this property. I leave this informal and incomplete hypothesis to future research.

3.6.5. Future research

Having presented several phase transformations to make formal Croft’s notions of scalar adjustment, many questions are left open. Indeed, much of the interest of a formal version of Croft’s system should lie in the classification and regimentation of those transformations. For instance, it is clear that for them to be well-defined, they should respect homogeneous subspaces. But what are the necessary conditions for a transformation to forget about a point? What transformations should count as equivalent, exactly?^{ax} Shifting operations seem to be available on the profile, in addition to the trimming operation Γ , as evidenced by alternations between inception and result state. Which are they?

^{ax}Just like for trajectories, domain and codomain self-equivalences could be used to define equivalence of transformations. For instance: two transformations $\tau_1, \tau_2 : X \rightarrow X'$ are equivalent iff there are two automorphisms $\varphi : X \rightarrow X$ and $\varphi' : X' \rightarrow X'$ such that $\tau_2 = \varphi' \circ \tau_1 \circ \varphi$. Or they could be said to be equivalent if their effects on equivalence classes of trajectories are identical.

4. Conclusion

In this thesis, I have proposed to make more precise Croft's (2012) system using topological tools. This is done without affecting his theoretical backdrop for construals or frame semantics, though with some changes in interpretation for his q dimension, which I now call phase spaces. Specifically, each aspectual type consists in three pieces of data. First, a phase space which registers all points that are aspectually relevant for the aspectual type, with an adequate notion of cohesion between them (a topology). Second, a trajectory within this phase space, modeled as a specific continuous function from a closed time interval to the phase space. Third, a specific **profiled** subpart of this trajectory, consisting of stages at, and transitions between, certain parts of the phase space which are called homogeneous subsets. These three elements make precise Croft's high-level description of aspect as **the two-dimensional unfolding of events over time**, and more specifically his preformal account of it as involving specific **sequences of qualitative states**. This sequence of qualitative states is not taken to be characteristic of each event type anymore, however.

Aspectual types can then be put in relation to one another thanks to phase transformations, which are essentially continuous functions between the phase spaces themselves. Depending on the operation, the transformation might forget about certain points of the input space, collapse certain distinctions inside the phase space, or make the space more cohesive. In each case, the coarse-grained construal corresponds to the output space of the phase transformation: these are always in a sense coarser than their input counterparts. Thus, topological spaces and maps between them can give us a precise notion of granularity, or more precisely relative granularity. Though by definition all phase transformations go from a more fine-grained space to a less fine-grained one, this does not mean that the input space should be the default one; the function is simply a given of aspectual competence, with no commitment as to how that function is used by the algorithms underlying aspectual performance. Interestingly, these phase transformations shift the focus away from scalar adjustment along time, and raises the possibility that scale in q could be just as or even more important than temporal scale for aspectual phenomena, contrary to most earlier accounts.

4.1. Changes in interpretation and classification

Meaning of q points This formal system is accompanied by a change in interpretation for the q dimension, which leads me to revise the aspectual classification slightly. Indeed, I propose that the formal representation of aspectual construals is indifferent to the meaning of the event types that are construed. Instead, points on the q dimension are distinguished only inasmuch as they matter for the unfolding of the event, and I eschew considerations on what their further meaning might be. Relatedly, while each phase space does register some flavor of qualitative change, there is no requirement that the change be in fact objectively measurable.

Incrementality and accomplishments As a consequence, since the kinds of incremental themes (mereological, property...) do not seem to matter aspectually, they are analyzed in the same way (just like in Croft 2012). The same goes for nonincremental accomplishments: motivated by the unnaturalness of their phase space, I put into question whether they should be distinguished from incremental accomplishments at all. For the resulting, unified type, I assume that the linear continuum of points is simply a given of our conceptual capacity, not a scale on which the entity can be placed precisely at any time point. However, I do analyze other aspectual types with sharp distinctions between topological spaces. Because of this, I argue contra Croft (2012) that the q states for undirected and directed (accomplishment) uses of motion verbs are crucially different, above and beyond the common nature of the motion. An undirected construal can be applied to a situation where the motion is directed; but

in that case the directedness is ignored, and this is reflected in the topological space.

Discontinuities and achievements Another consequence of this fact is that no phase space is assumed to be disconnected: “discontinuities” would only be warranted for states construed as permanent, but since the presence of inaccessible states is not relevant to an aspectual trajectory in itself, I analyze permanent spaces with a singleton phase space. In my system, Croft’s discontinuities actually involve two distinct states sitting immediately next to each other and being mutually accessible. When the change occurs, this does lead to intuitive discontinuity, but not topological discontinuity. Thus, I reject the aspectually irrelevant informal distinction between discontinuous/abstract conceptual spaces (*obtain*) and continuous/spatial ones (*arrive*). Similarly, I argue that a single space, whose active subset is uniform, can be used for undirected activities. Note that this is irrespective of whether an activity might involve cycling between two, three, or uncountably many states. Instead, I assume that a continuum is conceptually simpler; movement within this continuum, as opposed to stillness, then provides a nice notion of a minimal event.

Subtypes of achievements Finally, because it involves a perfectly symmetrical time interval, my system does not naturally allow for irreversible, one-way change. I argue that this is not a flaw, but simply that temporal conceptualization of events is not concerned with irreversibility, and is instead a matter of world knowledge and general reasoning. This lets us clean up this relatively idiosyncratic distinction.

For somewhat more down-to-earth reasons, I also discard the special class of cyclic achievements or semelfactives, because it involves quite a peculiar kind of trajectory: one that backtracks, and transitions instantly back to the initial state. While this could make the shift to undirected activity natural in Croft’s system (provided that type be made more precise), in my system this is no longer the case, and in fact the trajectory stands out among the rest of the canonical trajectories. To remedy this gap, I take inspiration from Rothstein (2008), who proposed instead that semelfactives could be a subclass of activities, but one with a special connection to telic predicates, in that their minimal events are naturally atomic. To reflect this in my analysis of semelfactives, I give a specific phase transformation relating a telic construal for a minimal event, and an undirected activity construal for its iteration. In fact, this sidestep could justify the seemingly comparable naturalness of single-event and iterated-event lexicalizations for semelfactives (Smith 1997, p. 30).

4.2. Further work

Constraining variation The phase spaces hypothesized here can mostly be characterized intensionally as exactly those spaces that satisfy certain conditions, formulated in Section 3.3.6: having a unique open point, and so on. Though these generalizations about phase spaces constitute a first step, more conditions are yet to be formulated to fully circumscribe the variations that are possible in the system. Similar constraints are imposed on trajectories and profiling, but these aspects are not fully regimented, in that the constrained variation hypothesized is not entirely accounted for by the constraints I formulate. This is very much needed to increase the explanatory power of the tools used and the overall system.

For instance, conditions are yet to be put on initial subsets and on homogeneous subsets; if different initial and homogeneous subsets were allowed, many differently-behaved phase spaces could exist in the system, and a cogent reason for their non-inclusion should be given. This is also the case of canonical trajectories which seem to form a quite limited class, even though many trajectories, even pathological, are possible in each phase space. The same is true of phase transformations: for instance, it is not clear exactly what points in an input phase space can be discarded in the cases where the

transformations are only partial functions. Finally, better operations should be given to account for the changes in profiling resulting from phase transformations; in particular, other uses for the operation Γ could help secure its status, and a similar operation, needed for semelfactives, is also lacking.

In this endeavor, it might plausibly be necessary to revise the aspectual classification more deeply, with more compelling formal arguments than the conjectural ones I have sketched here. One example might be the class of directed activities, whose specific aspectual behavior in the current version of the system only seems exceptional in that they constitute a result-less (and hence atelic atelic) equivalent of accomplishments. The aspectual distinction with undirected activities might benefit from additional motivations, or from being dropped entirely.

Aspectual verbs This regimenting problem for phase transformations is especially relevant to aspectual verbs. In particular, the selectional restrictions on aspectual verbs should ideally be derived from a uniform description of each aspectual verb's effect on an underspecified aspectual type, such that this mapping is ill-defined on aspectual types that cannot be modified by a given aspectual verb. Relatedly, it is as yet unclear what transformations or operations could be used to describe the operators *STOP* and *CONTINUE*; this asymmetry should ideally find justifications, or disappear altogether. In addition to this attainable goal, it might be possible to relate each aspectual verb's profiling effect with the more traditional semantic puzzle of predicting soft presuppositional triggers, such as Abrusán (2011) and Mandelkern (2016). Aspectual verbs constitute a canonical class of soft triggers, and this formal system could be a first stepping stone towards formulating an algorithmic, theoretically motivated account of their behavior.

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A. Topological appendix

A.1. Separation axioms

In this whole subsection, X is a topological space.

Definition A.1. Let x, y be points of X . The points x and y are **distinguishable** iff one of them has a neighborhood that does not contain the other:

$$[(\exists N \in \mathcal{N}_x) y \notin N] \vee [(\exists M \in \mathcal{N}_y) x \notin M].$$

Axiom T_0 . The space X is **T_0** iff any two distinct points are distinguishable:

$$(\forall x, y \in X) x \neq y \implies [(\exists N \in \mathcal{N}_x) y \notin N] \vee [(\exists M \in \mathcal{N}_y) x \notin M].$$

Definition A.2. Let x, y be points of X . The points x and y are **separated** iff each of them has a neighborhood that does not contain the other:

$$[(\exists N \in \mathcal{N}_x) y \notin N] \wedge [(\exists M \in \mathcal{N}_y) x \notin M].$$

Axiom R_0 . The space X is **R_0** (or **symmetric**) iff any two distinguishable points are separated:

$$(\forall x, y \in X) [(\exists N \in \mathcal{N}_x) y \notin N] \vee [(\exists M \in \mathcal{N}_y) x \notin M] \implies [(\exists N \in \mathcal{N}_x) y \notin N] \wedge [(\exists M \in \mathcal{N}_y) x \notin M].$$

Equivalently:

$$(\forall x, y \in X) [(\exists N \in \mathcal{N}_x) y \notin N] \implies [(\exists M \in \mathcal{N}_y) x \notin M].$$

Axiom T_1 . The space X is **T_1** (or **separated**) iff any two distinct points are separated:

$$(\forall x, y \in X) x \neq y \implies [(\exists N \in \mathcal{N}_x) y \notin N] \wedge [(\exists M \in \mathcal{N}_y) x \notin M].$$

Conversely, a space that does not satisfy this is called **non-separated**.

Axiom T_2 . The space X is **T_2** iff any two distinct points are separated by disjoint neighborhoods:

$$(\forall x, y \in X) x \neq y \implies [(\exists N \in \mathcal{N}_x) (\exists M \in \mathcal{N}_y) N \cap M = \emptyset].$$

A.2. Topology in terms of opens

Definition A.3. Given a set X with no structure, a **topology** $\mathcal{T}_X$ on this set is a collection of subsets of X , i.e. $\mathcal{T}_X \subseteq \mathcal{P}_X$,^{ay} that satisfies certain properties, given as Axioms O1–O3. The elements of $\mathcal{T}_X$ are called **open subsets of X** , or ‘opens’ for short.

Axiom O1. The total subset X is open (i.e. is an element of $\mathcal{T}_X$):

$$X \in \mathcal{T}_X.$$

Axiom O2. The union of an arbitrary collection of open subsets is still open:

$$(\forall A \text{ set}) (\forall (O_\alpha)_{\alpha \in A} : A \rightarrow \mathcal{T}_X) \bigcup_{\alpha \in A} O_\alpha \in \mathcal{T}_X.$$

^{ay}Since $\mathcal{P}_X$ is the collection of all subsets of X , writing ‘ $\mathcal{T}_X \subseteq \mathcal{P}_X$ ’ expresses that all elements of $\mathcal{T}_X$ are subsets of X . The converse may not be true, of course.

In particular, the empty subset $\emptyset$ is open (as the union of the empty collection of open subsets).

Axiom O3. *The intersection of any two opens is still open:*

$$(\forall O, O' \in \mathcal{T}_X) O \cap O' \in \mathcal{T}_X.$$

Once these axioms are in place, it is easy to see that there is no need for a counterpart to Axiom N2 for open sets. What's more, the fact that open sets are a more streamlined formulation of the notion of cohesion is transparent in the fact that Axiom N5, which was required for neighborhoods, is just a consequence of the definition of an open set as a set which neighbors all its points; the original axiom actually foreshadowed the notion of open set by appealing to the set $\{y \in X \mid N \in \mathcal{N}_y\}$.

Definition A.4. Let X be a set and $\mathcal{T}_X$ be a topology on this set. Let x be a point of X and N be a subset of X . The subset N is called a **neighborhood of x** iff N includes an open subset of $\mathcal{T}_X$ which also contains x :

$$(\exists O \in \mathcal{T}_X) O \subseteq N \wedge x \in O.$$

Definition A.5. Let X be a set and $\mathcal{T}_X$ be a topology on this set. Let $\mathcal{B}$ be a collection of open subsets of X (i.e., $\mathcal{B} \subseteq \mathcal{T}_X$). The collection $\mathcal{B}$ is called a **basis for the topology $\mathcal{T}_X$** iff for every open subset O and every point x within that subset, there is an element of $\mathcal{B}$ (a basic open) that contains x and is included in O :

$$(\forall O \in \mathcal{T}_X) (\forall x \in O) (\exists B \in \mathcal{B}) x \in B \wedge B \subseteq O.$$

Opens can be recovered from the basis by the following fact: any open subset is a union (possibly empty) of basic opens.

$$(\forall O \in \mathcal{T}_X) (\exists A \text{ set}) (\exists (B_\alpha)_{\alpha \in A} : A \rightarrow \mathcal{B}) O = \bigcup_{\alpha \in A} B_\alpha.$$

Proposition A.6. Let X be a space with a topology $\mathcal{T}_X$, and the corresponding collections of neighborhoods $(\mathcal{N}_x)_{x \in X}$. A subset F of X is closed in the sense above just in case any point all of whose neighborhoods meet F is in F , i.e.:

$$(\forall x \in X) [(\forall N \in \mathcal{N}_x) N \cap F \neq \emptyset] \implies x \in F.$$

Proof.

$$F \text{ is closed} \iff X \setminus F \text{ is open}$$

$$\iff (\forall x \in X) x \in X \setminus F \implies X \setminus F \in \mathcal{N}_x \quad \text{by Definition 2.15,}$$

$$\iff (\forall x \in X) \neg(x \in F) \implies ((\exists N \in \mathcal{N}_x) N \cap F = \emptyset) \quad \text{by Axiom N3 with } N \subseteq X \setminus F$$

$$\iff (\forall x \in X) [(\forall N \in \mathcal{N}_x) N \cap F \neq \emptyset] \implies x \in F \quad \text{by contraposition.}$$

□

Definition A.7. Let $\langle X, \mathcal{T}_X \rangle$ and $\langle Y, \mathcal{T}_Y \rangle$ be two topological spaces, and $f : X \rightarrow Y$ be a function between their underlying sets. The function f is **continuous** iff the preimage of any open in Y is an open in X :

$$(\forall O \in \mathcal{T}_Y) f^*(O) \in \mathcal{T}_X.$$

Proposition A.8. Let X, Y be two topological spaces and $f : X \rightarrow Y$ be a function between their underlying sets. The function f is continuous in the sense of Definition 2.13 if and only if it is continuous in the sense of Definition A.7.

Proof. $\Rightarrow$ Assume the function is continuous at all points $x \in X$ in the first sense and show that it is continuous in the second sense. For this, take any open set O of Y , which by assumption is a neighborhood of all its points, and show that $f^*(O) \subset X$ is also an open, i.e. is also a neighborhood of all its points. To check this, let x be a point in $f^*(O)$: because $f(x) \in O$ in Y by definition of the preimage, we have $O \in \mathcal{N}_{f(x)}$ (since O is a neighborhood of all its points). But then by continuity assumption this means $f^*(O) \subset X$ is a neighborhood of x . Since this is true of all points $x \in f^*(O)$, this means that $f^*(O)$ is an open set of X , thus proving the first implication.

$\Leftarrow$ Assume that the function is continuous in the second sense and show it is continuous at all points in the first sense. Take any point $x \in X$, and any neighborhood of its image $N \in \mathcal{N}_{f(x)}$ in Y . By equivalence between neighborhoods and opens, there is an open $O \subset Y$ included in N which contains $f(x)$. Since f is continuous in the first sense, this means that $f^*(O) \subset X$ is open. But since preimage is monotonous and $\{y\} \subseteq O \subseteq N$, we have $x \in f^*(\{y\}) \subseteq f^*(O) \subseteq f^*(N) \subset X$. Thus $f^*(N)$ contains an open which contains x , therefore is a neighborhood of x . This property being true for all x and all neighborhoods $N \in \mathcal{N}_{f(x)}$, this means that f is continuous at all points in the first sense. $\square$

A.3. Metric spaces

Definition A.9. Let X be a set, and d_X be a function $X \times X \rightarrow \mathbf{R}_+$: it associates a non-negative number to any ordered pair of points. The function d_X is called a **distance**, and the pair $\langle X, d_X \rangle$ is called a **metric space**, if the following conditions are met:

- **separation**—if two points are at distance zero, they are identical:

$$(\forall x, y \in X) \ d_X(x, y) = 0 \implies x = y;$$

- **symmetry**—the order of the points does not matter to compute their distance:

$$(\forall x, y \in X) \ d_X(x, y) = d_X(y, x);$$

- **triangle inequality**—“the straight path from x to z is always shorter or at worst as long as the path from x to z via y ” (nLab authors 2024):

$$(\forall x, y, z \in X) \ d_X(x, z) \leq d_X(x, y) + d_X(y, z).$$

In particular, two points that are close to a third one must be quite close to one another as well.

Definition A.10. Let d_X is a distance on X , y be a point of X and ε be a positive real number. The **open ball of radius ε around y** is defined as: $B(y, \varepsilon) := \{x \in X \mid d_X(y, x) < \varepsilon\}$.

Definition A.11. Let $\langle X, d_X \rangle$ be a metric space. Define **the induced topology on X** by taking as a neighborhood basis around any point x the set of open balls (Definition A.10) centered on x , i.e.:

$$(\forall x \in X) \ \mathcal{B}_x := \{B(x, \varepsilon) : \varepsilon > 0\}.$$

That is:

$$(\forall x \in X) \ \mathcal{N}_x := \{N \subseteq X \mid (\exists \varepsilon > 0) B(x, \varepsilon) \subseteq N\}.$$

Definition A.12. The topology $\mathcal{T}$ on a space X is called **metrizable** if there is a distance d on X such that $\mathcal{T}$ coincides with the topology induced by d .

Proposition A.13. Any metrizable topological space is T_2 .

Proof. Take a space $\langle X, \mathcal{T} \rangle$ and a distance d that induces the topology $\mathcal{T}$. Take two distinct points x and y in X : they lie a nonzero distance away from each other, say α with $\alpha > 0$ (otherwise they would not be distinct per the axiom of separation in Definition A.9). Then, $B(x, \alpha/4)$ is a neighborhood of x that does not contain y and conversely for $B(y, \alpha/4)$. Furthermore, these two balls are disjoint: if there was a point z belonging to both, this would violate the triangle inequality since we would derive $\alpha = d(x, y) \leq d(x, z) + d(z, y) < \alpha/2$. But $\alpha > 0$, so we in fact have $\alpha/2 < \alpha$, so the assumption that the balls have a common point is contradictory and they are indeed disjoint. This gives two disjoint neighborhoods of x and y respectively, as required for Axiom T_2 . $\square$

Proposition A.14. Let $\langle X, d_X \rangle$ and $\langle Y, d_Y \rangle$ be two metric space, $f : X \rightarrow Y$ be a function of sets between the two, and x be a point of X . Then the function f is $\varepsilon\delta$ -continuous at x iff it is (topologically) continuous at x .

Proof. Indeed, in a metric space, per Definition A.11, a neighborhood basis of a point z is given by the open balls of all radii around z (Definition A.10). Let us call $\mathcal{B}_z$ this neighborhood basis. Then the following equivalences naturally come as a consequence of this fact:

f is $\varepsilon\delta$ -continuous at x

$$\begin{aligned}
&\iff (\forall \varepsilon > 0) \quad (\exists \delta > 0) \quad (\forall x' \in X) \quad d_X(x', x) < \delta \implies d_Y(f(x'), f(x)) < \varepsilon && (\varepsilon\delta\text{-definition}) \\
&\iff (\forall N \in \mathcal{B}_{f(x)}) \quad (\exists \delta > 0) \quad (\forall x' \in X) \quad d_X(x', x) < \delta \implies f(x') \in N && (\text{definition of } \mathcal{B}_{f(x)}) \\
&\iff (\forall N \in \mathcal{N}_{f(x)}) \quad (\exists \delta > 0) \quad (\forall x' \in X) \quad d_X(x', x) < \delta \implies f(x') \in N && (\text{prop. is equiv. on neighb. basis}) \\
&\iff (\forall N \in \mathcal{N}_{f(x)}) \quad (\exists M \in \mathcal{B}_x) \quad (\forall x' \in X) \quad x' \in M \implies f(x') \in N && (\text{definition of } \mathcal{B}_x) \\
&\iff (\forall N \in \mathcal{N}_{f(x)}) \quad (\exists M \in \mathcal{B}_x) \quad M \subseteq f^*(N) && (\text{definition of preimage}) \\
&\iff (\forall N \in \mathcal{N}_{f(x)}) \quad f^*(N) \in \mathcal{N}_x && (\text{def. of neighborhood basis}).
\end{aligned}$$

$\square$

A.4. Another function $I \rightarrow S$

Define the function $p_{\text{accum}} : I \rightarrow S$ by:

$$t \in I \mapsto \begin{cases} c & \text{if } t = \frac{2}{3} \text{ or } (\exists n \geq 1) t = \frac{2}{3} - \frac{1}{3n} \\ g & \text{otherwise.} \end{cases}$$

Then this function is continuous, because $p_{\text{accum}}^*(\{g\})$ is the following union of open intervals:

$$[0, \frac{1}{3}) \cup (\frac{2}{3}, 1] \cup \bigcup_{n \geq 1} (\frac{2}{3} - \frac{1}{3n}, \frac{2}{3} - \frac{1}{3(n+1)}).$$

This union is clearly a neighborhood of all its points in $[0, 1]$.

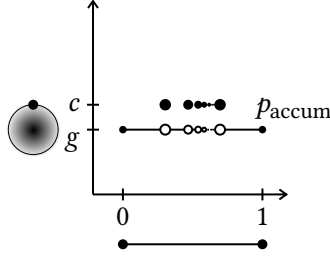


Figure 30: The continuous trajectory p_{accum}

A.5. Equivalence classes and quotients

Definition A.15. Let X be a topological space and R be an equivalence relation on X . The **quotient of X by R** is the set of equivalence classes of X , notated X/R .

Let $\chi \in X/R$ and a subset $N \subseteq X/R$. The **topology on X/R** is given by the following condition. The subset N is a neighborhood of χ iff for all $x \in \chi$ (i.e. all x which are members of the equivalence class) the subset $\bigcup_{\xi \in N} \xi$ (i.e. the union of all equivalence classes which constitute N) is a neighborhood of x :

$$(\forall x \in \chi) \bigcup_{\xi \in N} \xi \in \mathcal{N}_x.$$

Proposition A.16. For each $x \in X$, let $[x]$ denote the equivalence class of x under R . With the topology above, the quotient map $\pi_X^R : X \rightarrow X/R, x \mapsto [x]$ is also continuous.

A.6. Manifolds

Definition A.17. Let $|\cdot|$ be the function $\mathbf{R}^n \rightarrow \mathbf{R}, (x_1, \dots, x_n) \mapsto \sum_i x_i^2$ (the Euclidean norm squared). The **open ball of dimension n** , notated $B_{\mathbf{R}^n}$, is the set of n -dimensional vectors which are sent to a number smaller than 1 by $|\cdot|$:

$$\begin{aligned} B_{\mathbf{R}^n} &:= \{(x_1, \dots, x_n) \in \mathbf{R}^n \mid \sum_i x_i^2 < 1\} \\ &= |\cdot|^{-1}([0, 1)). \end{aligned}$$

This is the same as the open ball of radius 1 around $(0, \dots, 0)$ for the Euclidean distance.

Definition A.18. The **open half-ball of dimension n** , notated $B_{\mathbf{R}^n}^+$, is the set of n -dimensional vectors that are inside $B_{\mathbf{R}^n}$ and whose first coordinate is nonnegative:

$$\begin{aligned} B_{\mathbf{R}^n}^+ &:= B_{\mathbf{R}^n} \cap \{(x_1, \dots, x_n) \in \mathbf{R}^n \mid x_1 \geq 0\} \\ &= \{(x_1, \dots, x_n) \in \mathbf{R}^n \mid x_1 \geq 0 \wedge \sum_i x_i^2 < 1\}. \end{aligned}$$

Its **bottom surface** is notated $B_{\mathbf{R}^n}^0$ and defined as:

$$B_{\mathbf{R}^n}^0 := \{(x_1, \dots, x_n) \in \mathbf{R}^n \mid x_1 = 0 \wedge \sum_i x_i^2 < 1\}.$$

Definition A.19. Let X be a topological space, ∂X be a subset of X , and n be a natural number. The space X is a **manifold of dimension n with boundary ∂X** iff:

- the subset $x \in X \setminus \partial X$ is a manifold;

- for every point $x \in \partial X$, there is a neighborhood N of that point which is homeomorphic to the open half-ball $B_{\mathbf{R}^n}^+$ of dimension n , through a homeomorphism which sends all points in the boundary ∂X to the bottom surface of the ball:

$$(\forall x \in \partial X) (\exists N \in \mathcal{N}_x) (\exists \varphi : N \rightarrow B_{\mathbf{R}^n}) (\exists \psi : B_{\mathbf{R}^n} \rightarrow N)$$

$$\psi \circ \varphi = \text{id}_N \wedge \varphi \circ \psi = \text{id} \wedge \varphi^* (B_{\mathbf{R}^n}^0) = N \cap \partial X.$$

B. Phase spaces

Aspectual type	Phase space	Homogeneous spaces		
Permanent state	P	$\{g_P\}$		
Transitory state	S	$\{g\}$		$\{c\}$
Achievement (directed)	S	$\{g\}$		$\{c\}$
Accomplishment	I	$\{g_I\}$	$[0_I, 1_I)$	$\{1_I\}$
Directed activity	D	$\{g_D\}$	$[0_D, 1_D)$	
Undirected activity	U	$\{g_U\}$	$(0_U, 1_U)$	

Table 3: Phase space and homogeneous spaces for each aspectual type